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TOPICAL REVIEW

Graph Neural Networks for Evaluating the Reliability and Resilience of Infrastructure Systems: A Systematic Review of Models, Applications, and Future Directions

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ABSTRACT Interdependent infrastructure systems serve as the foundation of modern society, ensuring the delivery of essential services, economic stability, and public well-being. The increasing integration of these systems, driven by advancements in cyber-physical systems (CPS) and the Internet of Things (IoT), has improved operational efficiency and adaptability. However, this interconnectivity and interdependency have also introduced vulnerabilities, making infrastructure networks more susceptible to cascading failures, cyber attacks, and disruptions from natural disasters and human-induced events. Graph neural networks (GNNs) have emerged as a powerful tool for addressing these challenges, offering data-driven solutions for reliability and resilience analysis across interconnected infrastructure networks. This paper provides a comprehensive review of GNN applications in interdependent infrastructure systems, including transportation networks, power distribution networks, water distribution networks, and communication networks. By leveraging spatial-temporal modeling, multi-modal data integration, and physics-informed learning, GNN-based approaches enhance predictive accuracy, system resilience, and decision-making efficiency. This review underscores the potential of GNNs in infrastructure management optimization and risk mitigation, and enabling the development of more sustainable, adaptive, and resilient infrastructure systems in the face of evolving challenges.

INDEX TERMS Graph neural network, critical infrastructure systems, reliability analysis, resilience analysis, interdependency.

I. INTRODUCTION

A. IMPORTANCE OF INFRASTRUCTURE SYSTEMS

Infrastructure systems build the foundation for the functionality and sustainability of modern societies, encompassing various networks such as transportation systems, water distribution networks (WDNs), power distribution networks (PDNs), and communication networks [1], [2], [3], [4]. These systems are essential for ensuring the delivery of basic services, maintaining economic stability, and supporting well-being. Their efficient management and operation are critical for public health, safety, and overall quality of life. For example, WDNs are designed to deliver clean and safe water to all residents, meeting essential needs such as drinking, sanitation, and industrial use [2], [3]. Transportation systems ensure the mobility of goods and services. They connect cities and regions, enabling economic activity and access to essential resources. Similarly, PDNs are pivotal in sustaining the operation of other critical systems, including water supply, transportation, and telecommunications [5].

In addition to the functionality of each standalone infrastructure system, different systems often interact with and depend on one another. Specifically, the advent of the Internet of Things (IoT) and cyber-physical system

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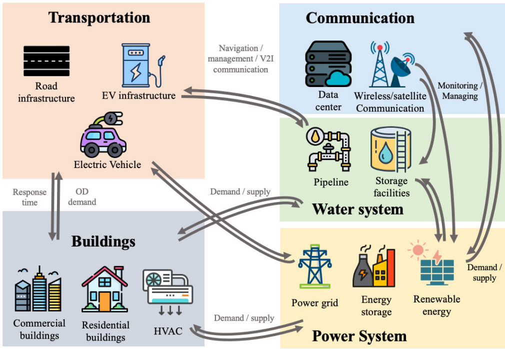

FIGURE 1. Interdependencies among infrastructure systems highlighting key connections between transportation, communication, water systems, buildings, and the power system. The arrows represent demand, supply, and information exchange, such as electric vehicle reliance on the power grid, navigation and communication support, and water system dependencies on energy supply.

(CPS) have revolutionized traditional infrastructure into integrated, intelligent systems [6], [7], [8]. IoT networks incorporate physical components like sensors and gateways, which collect and transmit data through the communication systems in real-time [9]. Such systems can identify anomalies, optimize resource usage, and even predict potential failures, enabling infrastructure operators to respond swiftly and effectively. As illustrated in Figure 1, a concrete example of this integration can be seen in IoT-enabled smart grids, where sensors monitor electricity usage, grid stability, and environmental conditions. Similarly, intelligent transportation systems leverage GPS devices, road cameras, and smart traffic signals to collect and analyze traffic flow data. These advancements have transformed infrastructure systems into adaptive and intelligent networks capable of responding efficiently to various demands.

However, these infrastructure systems are vulnerable to a variety of scenarios, including human-induced incidents such as traffic accidents, as well as natural hazards like extreme rainfall, storm surges, and floods [10], [11], [12], [13]. Furthermore, for interconnected infrastructure systems, these events often trigger cascading failures, where disruptions in one system propagate to others, amplifying the overall impact and posing significant risks to communities and economies [14]. For instance, when the city is impacted by floods or fires, transportation networks should be able to facilitate the evacuation of affected populations and ensure the delivery of emergency supplies. Road and railway systems must remain operational to maintain accessibility for the public and support disaster response efforts [15], [16]. By addressing these vulnerabilities, infrastructure systems can reduce the likelihood of cascading failures and mitigate the adverse effects of both human-induced and natural disruptions. These advancements are essential to ensuring the continued functionality and sustainability of critical infrastructure in the face of evolving challenges.

To evaluate the quality of infrastructure conditions, service capabilities, and maintenance functionality under adverse events, reliability and resilience analysis of regional infrastructure systems has become increasingly critical [17]. This analysis is especially important given the pivotal role these systems play in maintaining essential services during natural disasters and adapting to the pressures of rapid urbanization. Each type of infrastructure requires specific metrics for evaluation, tailored to its unique functionality and vulnerabilities. For example, in transportation systems, quantifying accessibility under natural hazards is crucial. Accessibility metrics such as travel time [18], connectivity [19], and evacuation efficiency [20] are often used to assess the system's ability to maintain operations during floods, earthquakes, or other extreme events. In PDNs, the stability of voltage [21] is a key performance indicator, as fluctuations or failures can disrupt electricity supply to critical services, such as hospitals and emergency response facilities. Similarly, in WDNs, maintaining water quality is a vital metric. This includes monitoring contamination levels [22], flow rates [23], and pressure level [24] to ensure a reliable and safe water supply during adverse events, such as flooding or infrastructure failures. In communication systems, resilience to cyber-physical attacks is a critical concern [25]. Metrics such as latency, data integrity, and network availability help quantify the system's ability to withstand and recover from attacks that could disrupt essential communication services. By employing domain-specific metrics for reliability and resilience analysis, infrastructure systems can be better evaluated and improved to withstand adverse events. These assessments enable stakeholders to identify vulnerabilities, prioritize upgrades, and implement proactive measures to ensure that infrastructure systems remain functional and resilient, even in the face of evolving challenges.

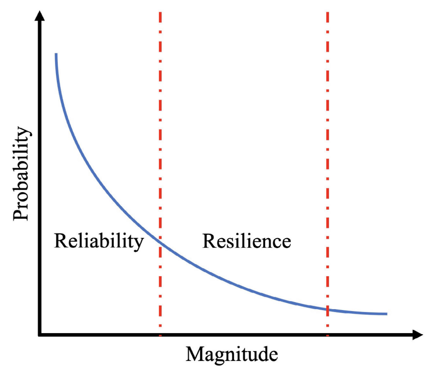

FIGURE 2. Probability-Magnitude Relationship in Reliability and Resilience.

B. RELIABILITY AND RESILIENCE IN INFRASTRUCTURE SYSTEM

Reliability and resilience are distinct yet interconnected concepts, as illustrated in Figure 2 and 3. Reliability refers to a system's ability to perform its intended functions under specified conditions without failure [26]. It focuses

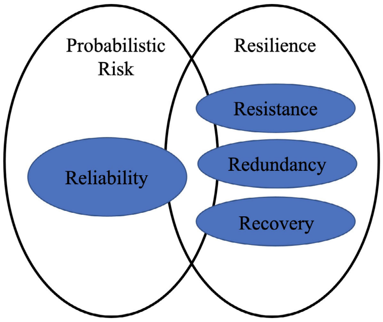

FIGURE 3. Framework of Probabilistic Risk, Reliability, and Resilience Components.

on consistent and dependable performance during normal operations. In contrast, resilience encompasses a broader range of attributes including reliability. It emphasizes a system's capacity to withstand, adapt to, and recover from disruptions. Resilience can be further interpreted as a system's ability to mitigate the impact of the hazards [27], maintain essential functions during crises, and leverage available resources to enhance its performance post-crisis. The resilient infrastructure systems should consist of four components [28], [29]: resistance, reliability, redundancy, and recovery, as illustrated in Figure 4.

- Resistance: the ability to prevent damage or disruption by providing protection against hazards and mitigating their impacts [30]. For instance, flood barriers in WDNs help reduce the impact of storm surges and heavy rainfall.
- Reliability: the capacity to operate effectively under a wide range of conditions [29]. For example, PDNs should be able to maintain voltage stability under fluctuating demands.
- Redundancy: the redundancy ensures that infrastructure systems have sufficient backup capacity to enable functionality [28]. For instance, transportation systems with alternative routes and bypasses enable uninterrupted mobility when primary roads are blocked or congested.
- Recovery: the ability to respond and recover swiftly and efficiently from disruptive events [28]. For instance, a communication network could reconfigure rapidly to reroute data during a cyberattack.

While reliability focuses on minimizing failures during normal operations, resilience emphasizes a system's ability to handle and recover from abnormal conditions. Both concepts are essential for comprehensive infrastructure management, as they ensure continuous functionality and adaptability to evolving challenges. By integrating reliability and resilience metrics, decision-makers can better evaluate system performance and implement targeted policies to strengthen infrastructure networks.

C. RELIABILITY AND RESILIENCE ANALYSIS OF INFRASTRUCTURE SYSTEMS

A wide range of approaches has been proposed for the reliability analysis of infrastructure systems. These methods can be broadly categorized into four major types:

- **Deterministic Approaches** Deterministic methods focus on analyzing infrastructure systems under fixed conditions and predefined parameters. These approaches typically use mathematical models to evaluate system performance and identify potential points of failure [32], [33]. For instance, [34] proposes a deterministic network flow model to assess the reliability of transportation systems by identifying critical links or nodes that could cause disruptions.
- **Probabilistic Approaches** Probabilistic methods incorporate uncertainties into reliability analysis by using statistical models to evaluate the likelihood of failures. These approaches account for variability in factors such as demand fluctuations, environmental conditions, and component aging [35], [36], [37]. For instance, [38] conducted probabilistic load flow analysis on power distribution networks to estimate voltage stability under uncertain load conditions. Furthermore, Bayesian-based methods have also been employed to determine the probability distributions of infrastructure failure scenarios, offering insights into system vulnerabilities under uncertain conditions [39], [40].
- **Simulation-based Approaches** Simulation-based techniques use computational models to mimic the behavior of infrastructure systems under various scenarios [41], [42]. These methods are particularly useful for analyzing complex systems with dynamic interdependencies. For instance, Monte Carlo simulations are frequently employed to assess the reliability of water distribution systems by simulating pipe failures and their cascading effects on the network [43].
- **Machine Learning-based Approaches** Machine learning and artificial intelligence have gained significant attention in reliability analysis in infrastructure systems [44]. These methods use historical data and predictive models to identify failure patterns and predict system performance. For instance, fully connected neural networks (FCNNs) [45], convolutional neural networks (CNNs) [46], [47], and recurrent neural networks (RNNs) [48], [49] have been utilized to analyze time-series data, such as traffic patterns and electricity demand, for anomaly detection and predictive maintenance. However, conventional neural networks face limitations, including challenges with generalization across different topologies and difficulty integrating heterogeneous, multi-modal data sources.

Among existing approaches, machine learning-based methods have emerged as transformative tools for infrastructure reliability analysis. However, conventional neural networks like FCNN, CNN, and RNN often struggle to handle non-Euclidean data structures, such as transportation networks or power grids, and lack the flexibility to generalize across varying topologies [50].

Graph Neural Networks (GNNs) have recently gained significant attention as a promising solution to these challenges. By leveraging their ability to model graph-

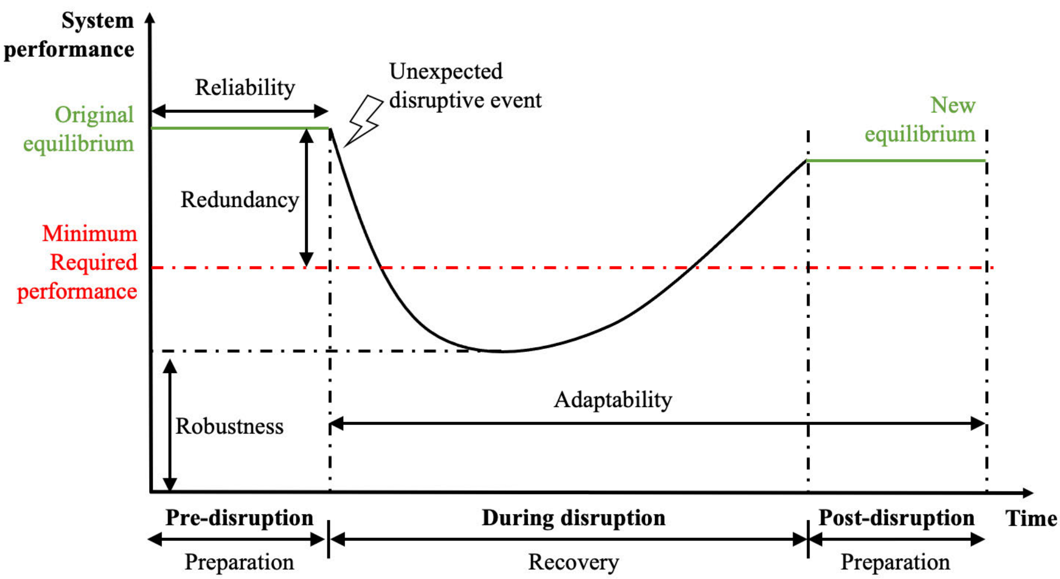

FIGURE 4. Schematic of performance of a resilient system (adapted from [31]).

structured data, GNNs excel in capturing relationships and dependencies within infrastructure systems. Unlike traditional neural networks, GNNs utilize message-passing mechanisms to encode spatial interactions between nodes and their neighbors. This capability allows GNNs to handle unstructured, dynamic data and generalize effectively to unseen scenarios. As a result, GNNs are well-suited for applications in infrastructure systems, particularly for modeling topological interdependencies, analyzing cascading failures, and supporting decision-making under uncertainty. However, the application of GNNs to reliability and resilience analysis is still an emerging field. There are several open questions and challenges that need to be addressed to fully explore their potentials:

- How can reliability analysis be effectively formulated as graph-based problems to leverage the strengths of GNNs?
- How can heterogeneous data sources (e.g., geographic, temporal, and domain-specific data) be integrated within a unified GNN framework to capture multi-modal relationships?
- Can GNN models generalize across different hazards, geographic regions, and infrastructure systems without requiring extensive finetuning or retraining?
- Are GNNs capable of processing data in real time to support timely decision-making for regional infrastructure management during crises?

To address these challenges, this paper offers a comprehensive review of GNN-based approaches for reliability and resilience analysis in infrastructure systems. By synthesizing existing literature and presenting a synergistic framework, this work provides valuable insights and guidelines for researchers and practitioners, particularly those new to the field. Ultimately, this paper aims to underscore the transformative potential of GNNs in improving the resilience and reliability of infrastructure systems, paving the way for more adaptive and sustainable solutions.

II. SURVEY ORGANIZATION

A. LITERATURE SELECTION CRITERIA

To ensure a comprehensive yet focused survey, the literature reviewed in this paper was selected based on three main criteria: (i) Relevance to infrastructure systems, with emphasis on transportation, water distribution, power distribution, and communication networks; (ii) Integration of graph neural networks into reliability or resilience analysis, including works that extend conventional machine learning to graph-based domains; and (iii) Scholarly quality and impact, prioritizing peer-reviewed journal articles, and conference proceedings. In addition, references were gathered through a combination of systematic keyword searches (e.g., "graph neural networks for infrastructure reliability," "resilient infrastructure GNN," and "GNN cascading failures"). The final corpus includes 657 papers spanning diverse application domains, and we keep 135 reference after screening. Among them, approximately 40% focus on transportation systems, reflecting the substantial research interest in traffic flow prediction, route optimization, and resilience assessment. Around 26% of the surveyed works are dedicated to power distribution networks, primarily addressing power flow optimization, fault detection, and cascading failure mitigation. Water distribution networks represent roughly 18% of the reference, with research concentrated on state estimation, contamination source detection, and sensor placement. The remaining 16% are centered on communication networks, where GNNs are employed for link scheduling, anomaly detection, and improving cyber-physical resilience. In terms

TABLE 1. Demographic summary of the surveyed literature.

Category	Count	Share (%)
Application domain		
Transportation networks	54	40%
Power distribution networks	35	26%
Water distribution networks	25	18%
Communication networks	21	16%
Publication year		
Before 2019	14	10%
2020 - 2022	46	34%
After 2023	75	56%
Total	135	100%

of publication trends, the literature demonstrates rapid growth since 2020, aligning with the increasing maturity of GNN frameworks and heightened interest in resilient infrastructure systems. More than 70% of the reviewed works were published between 2021 and 2025, indicating that this is an emerging research area with significant momentum. This demographic summary underscores the growing convergence between infrastructure engineering and machine learning communities, highlighting the need for integrative surveys such as the present one.

B. TAXONOMY

This section provides a taxonomy of GNNs for the reliability and resilience analysis of infrastructure systems, which also serves as a generalization of the subsequent content in this paper. As illustrated in Figure 5, our survey is organized into three main components that are critical for understanding the role of GNNs in this domain: the background of GNNs, their applications in infrastructure systems, and directions for future research.

We begin with an overview of GNN architectures, tasks, and formulations in Section III. This section provides a foundational understanding of GNN principles. In Section IV, we explore the application of GNNs to infrastructure systems, focusing on four distinct subfields:

- **Transportation networks (Section IV-A)** We highlight how GNNs are used for traffic flow prediction, route optimization, anomaly detection, and criticality analysis to improve transportation system reliability and efficiency.
- **Water distribution networks (Section IV-B)** This subsection reviews GNN applications for tasks such as state estimation, sensor placement optimization, and contamination source identification.
- **Power distribution networks (Section IV-C)** We discuss the use of GNNs in power flow optimization, fault detection, cascading failure mitigation, and resilience analytics.
- **Communication networks (Section IV-D)** We examine how GNNs are applied to optimize link scheduling, enhance network reliability, and detect anomalies in communication systems.

Following this, Section IV-E introduces GNN applications that account for the interdependencies between infrastructure systems. Subsequently, in Section V, we examine the challenges and opportunities for future applications of GNNs in infrastructure systems. Finally, we conclude and summarize the key findings in Section VI, providing a comprehensive outlook on the transformative potential of GNNs in enhancing the reliability and resilience of modern infrastructure systems.

III. GRAPH NEURAL NETWORKS

Compared with conventional neural networks, GNN is a powerful framework for analyzing graph-structured data, enabling the modeling of both spatial and temporal relationships [51]. By leveraging the inherent connectivity of graph data, GNNs can learn representations that effectively encode complex interactions. This section provides a comprehensive mathematical formulation of GNNs, encompassing both spatial and spatial-temporal GNNs.

A. SPATIAL GNNS

The general idea of GNNs is to utilize the structural information of graphs to perform node, edge, or graph-level predictions. A key operation in GNNs is called message passing, which allows nodes to share and update information based on their neighbors.

Spatial GNNs operate on static graph structures to aggregate and update node features based on the features of neighboring nodes. Let the graph be represented as $G = (V, E)$, where V is the set of nodes, and E is the set of edges. Each node $i \in V$ is associated with a feature vector $\boldsymbol{x}_i^{(l)}$ at layer l. A generic update rule for spatial GNNs can be expressed as:

$$\boldsymbol{x}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \phi\left(\boldsymbol{x}_i^{(l)}, \boldsymbol{x}_j^{(l)}, \boldsymbol{e}_{ij}\right)\right),$$

where $\mathcal{N}(i)$ denotes the set of neighbors of node i, $\boldsymbol{e}_{ij}$ represents the edge features between nodes i and j, ϕ is a learnable function, such as a multilayer perceptron (MLP), and σ is an activation function, e.g., ReLU or sigmoid [52].

Common spatial GNN architectures include Graph Convolutional Networks [53] (GCNs), Graph Attention Networks [54] (GATs), GraphSAGE [55]. These architectures differ primarily in how ϕ is defined and the aggregation mechanisms employed.

1) GRAPH CONVOLUTIONAL NETWORKS (GCNS)

Graph Convolutional Networks (GCNs) simplify the aggregation process by employing a spectral-based approach to aggregate neighboring node features. GCNs leverage spectral graph theory to propagate information through the graph, using convolutional operators to aggregate neighborhood features. A typical GCN update rule is:

$$\boldsymbol{X}^{(l+1)} = \sigma\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\boldsymbol{X}^{(l)}\boldsymbol{W}^{(l)}\right), \quad (1)$$

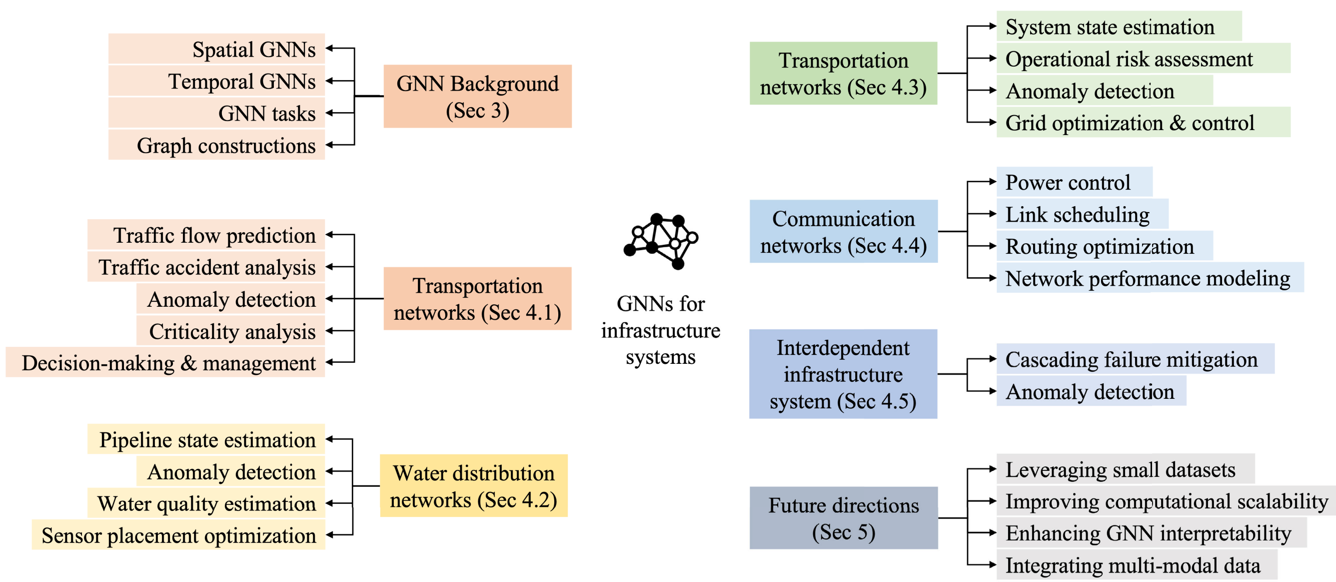

FIGURE 5. **The structure and outlines of the paper.**

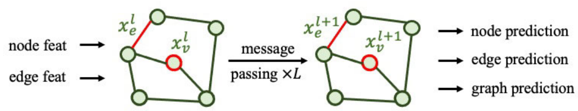

FIGURE 6. **General formulation of Graph neural network and message passing.**

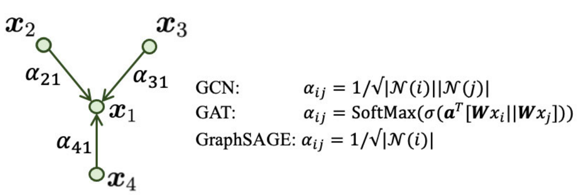

FIGURE 7. **General formulation of Graph neural network and message passing.**

where $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}$ is the adjacency matrix with added self-loops, $\tilde{\boldsymbol{D}}$ is the diagonal degree matrix of $\tilde{\boldsymbol{A}}$, $\boldsymbol{X}^{(l)}$ is the node feature matrix at layer l, and $\boldsymbol{W}^{(l)}$ is the learnable weight matrix. GCNs are computationally efficient and well-suited for tasks where global graph smoothness is important, such as node classification and community detection.

2) GRAPH ATTENTION NETWORKS (GATS)

Compared with GCNs using adjacency matrix, GATs introduce attention mechanisms to dynamically weight the importance of neighboring nodes. The attention coefficient for nodes i and j is calculated as:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}(\boldsymbol{a}^T[\boldsymbol{W}\boldsymbol{x}_i \| \boldsymbol{W}\boldsymbol{x}_j])\right)}{\sum_{k\in\mathcal{N}(i)} \exp\left(\text{LeakyReLU}(\boldsymbol{a}^T[\boldsymbol{W}\boldsymbol{x}_i \| \boldsymbol{W}\boldsymbol{X}_k])\right)},$$

where $\|$ denotes concatenation, and $\boldsymbol{a}$ and $\boldsymbol{W}$ are learnable parameters. GATs excel in capturing heterogeneity in node relationships, making them particularly useful for tasks with diverse and noisy graph structures.

3) GRAPHSAGE

GraphSAGE aggregates information by sampling a fixed-size set of neighbors and applying learnable aggregation functions. For node i, the aggregation is:

$$\boldsymbol{x}_i^{(l+1)} = \sigma\left(\boldsymbol{W}\left[\boldsymbol{x}_i^{(l)} \| \texttt{AGGREGATE}(\{\boldsymbol{x}_j^{(l)} : j \in \mathcal{N}(i)\})\right]\right), \quad (2)$$

where AGGREGATE is a pooling function, such as mean and max function. GraphSAGE samples a fixed number of neighbors, enabling scalability to large graphs. GraphSAGE is scalable and supports inductive learning, making it suitable for dynamic or partially observed graphs. These three models can be all represented with a attention coefficient, but with different expression, as shown in Figure 7.

B. SPATIAL-TEMPORAL GNNS

Spatial-temporal GNNs extend spatial GNNs by incorporating temporal dynamics into the graph structure. Let $\{G_t\}_{t=1}^T$ denote a sequence of graphs over T time steps. For each graph G_t, nodes and edges are associated with time-dependent features $\boldsymbol{x}_i^{(l,t)}$ and $\boldsymbol{e}_{ij}^{(t)}$, respectively.

A widely adopted update rule for spatial-temporal GNNs is:

$$\boldsymbol{x}_i^{(l+1,t)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \phi\left(\boldsymbol{x}_i^{(l,t)}, \boldsymbol{x}_j^{(l,t)}, \boldsymbol{e}_{ij}^{(t)}\right) + \psi\left(\boldsymbol{x}_i^{(l,t-1)}\right)\right),$$

where ψ is a temporal function, such as LSTMs [56] or GRUs [57], which model sequential dependencies.

Additionally, attention mechanisms can model spatial and temporal dependencies simultaneously. For instance, spatiotemporal attention computes weights over both spatial neighbors and past time steps to dynamically prioritize information flow. Spatial-temporal GNNs are particularly effective for analyzing dynamic systems where spatial and temporal interactions are interdependent.

In conclusion, the integration of temporal modeling into GNNs allows for a unified framework capable of capturing complex interactions in time-evolving systems, paving the way for advancements in regional resilience studies and related fields.

C. GNN TASKS

GNNs could be used to address a wide range of problems in graph-structured data. Broadly, these problems can be categorized into three primary tasks: node-level, edge-level, and graph-level tasks. Each category leverages the expressive power of GNNs to extract insights from graph representations.

1) NODE-LEVEL TASKS

Node-level tasks aim to infer properties or predict labels associated with individual nodes in a graph. These tasks typically involve aggregating information from neighboring nodes to update the feature representation of a target node. Mathematically speaking, the node regression task could be represented as

$$\boldsymbol{\wp}_i = \sigma\left(\sum_{j\in\mathcal{N}(i)} \phi\left(\boldsymbol{x}_i^{(l)}, \boldsymbol{x}_j^{(l)}, \boldsymbol{e}_{ij}\right)\right), \tag{3}$$

where $\boldsymbol{x}_i^{(l)}$ represents the feature vector of node i at layer l, $\mathcal{N}(i)$ denotes the set of neighbors of node i, $\boldsymbol{e}_{ij}$ captures the edge attributes between nodes i and j, ϕ is the message-passing function, and σ is a non-linear activation function. Typical applications include node classification (e.g., predicting the type of a user in a social network) and node regression (e.g., forecasting traffic flow at specific sensors in a transportation network).

2) EDGE-LEVEL TASKS

Edge-level tasks focus on predicting properties or labels of edges, which represent relationships or interactions between nodes. These tasks often involve aggregating node and edge features to infer the characteristics of a particular edge. The edge-level prediction can be expressed as:

$$\hat{\boldsymbol{y}}_{ij} = \psi\left(\boldsymbol{x}_i^{(l)}, \boldsymbol{x}_j^{(l)}, \boldsymbol{e}_{ij}\right), \tag{4}$$

where $\hat{\boldsymbol{y}}_{ij}$ is the predicted property or label of the edge connecting nodes i and j, and ψ is a function that combines node and edge features. Applications include edge classification (e.g., identifying critical links in a transportation network) and edge regression (e.g., estimating traffic flow between two locations).

3) GRAPH-LEVEL TASKS

Graph-level tasks aim to predict properties or labels associated with the entire graph. These tasks aggregate node and edge features across the graph to generate a holistic representation. The graph-level representation $\hat{\boldsymbol{h}}_G$ can be computed as:

$$\hat{\boldsymbol{h}}_G = \rho\left(\{\boldsymbol{x}_i^{(l)} | i \in \mathcal{V}\}, \{\boldsymbol{e}_{ij} | (i,j) \in \mathcal{E}\}\right), \tag{5}$$

where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of nodes and edges in the graph, respectively, and ρ is a graph-level pooling function, such as summation, mean, or max pooling. Typical applications include graph classification (e.g., determining molecular properties in a chemical graph) and graph regression (e.g., predicting the structural robustness of a transportation network).

The flexibility of GNNs across node-level, edge-level, and graph-level tasks underscores their adaptability to diverse problem settings. By leveraging tailored message-passing schemes and aggregation functions, GNNs can effectively model complex relationships in graph data, enabling breakthroughs in areas ranging from social networks to transportation systems.

D. GRAPH CONSTRUCTION IN INFRASTRUCTURE SYSTEMS

Graphs can be formulated in two distinct ways for infrastructure systems: one is based on sensor data and other installations within the infrastructure, tailored for specific tasks; the second method utilizes the proximity topology of the infrastructure itself. These formulations cater to diverse applications and system requirements, offering flexibility in how infrastructure networks are modeled and analyzed [51].

1) SENSOR-BASED GRAPH

The first approach involves constructing a graph based on sensors or other monitoring installations embedded within the infrastructure [51]. This formulation varies significantly across different tasks and infrastructure types due to differences in sensor characteristics, such as sensor types (e.g., pressure sensors, voltage monitors, or traffic counters), installation locations, and the extent of knowledge or coverage (full or partial) provided by the sensors [58]. For example, in water distribution networks, sensors may measure flow rates or pressure levels at critical junctions [59], while in power distribution systems, sensors track voltage stability and current flows at key substations [60]. As a result, the resulting graph topology is directly influenced by the spatial distribution and density of sensor placements. This sensor-based approach provides a highly task-specific framework for studying system dynamics and detecting anomalies or disruptions.

$$a_{ij}^t = \begin{cases} 1, & \text{if } v_i \text{ connects to } v_j, \\ 0, & \text{otherwise,} \end{cases} \tag{6}$$

where a_{ij}^t denotes an element in the adjacency matrix at time t, i and j are indices of vertices in the graph.

2) DISTANCE-BASED GRAPH

The second approach involves directly leveraging the inherent topology of the infrastructure system, independent from sensor locations. This formulation emphasizes modeling the physical or functional structure of the system itself, making it particularly effective for tasks such as resilience analysis, flow optimization, and criticality assessment [51], [61]. By focusing on the system's topology, this approach captures the relationships and interactions among its components, providing a deeper understanding of its behavior under various conditions.

For instance, in transportation networks, a graph representation can model roads as edges and intersections as nodes [62], [63]. This structure captures the system's connectivity and enables analyses such as traffic flow optimization, vulnerability assessments, and evacuation planning during emergencies. In power distribution systems, the graph may represent substations and transformers as nodes, with transmission lines as edges [64]. This representation facilitates the study of cascading failures, power flow optimization, and system resilience against natural disasters or cyberattacks. Similarly, in communication networks, the topology can model routers, switches, and data links, enabling the analysis of network latency, data flow optimization, and fault-tolerant routing [65].

To construct such graphs, the adjacency matrix is typically calculated using a kernel function that incorporates distances or other relevant physical attributes. Gaussian radial basis function and inverted function are two common kernel functions used in previous literature. For example, the adjacency matrix of a distance-based graph with Gaussian radial basis can be computed as:

$$a_{ij}^{t} = \begin{cases} \exp\left(-\dfrac{\|d_{ij}^{t}\|_2}{\sigma}\right), & \text{if } d_{ij}^{t} < \epsilon, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

where d_{ij}^{t} denotes the distance between node i and node j at time t; ϵ is a predefined threshold to control the sparsity of the adjacency matrix; σ is a hyper-parameter to control the distribution.

This comprehensive approach to graph construction enables the modeling of complex interdependencies and dynamic behaviors in power distribution systems, supporting advanced analytics such as anomaly detection, cascading failure analysis, and resilience assessment.

IV. GNN APPLICATIONS IN INFRASTRUCTURE SYSTEMS

A. TRANSPORTATION NETWORKS

Transportation networks rely heavily on reliability metrics to ensure consistent mobility, particularly during emergencies. Resilience evaluations assist in planning for disruptions such as bridge collapses, road blockages, or severe weather events. Applications include assessing travel time reliability, optimizing route planning, and developing contingency strategies to maintain accessibility and connectivity. Key components of infrastructure systems, such as bridges, roads, and highway networks, must be assessed comprehensively before, during, and after natural disasters. Commonly used metrics include accessibility [66], travel time [18], and resilience indices [67]. Recent advancements in graph neural networks have significantly enhanced the methodologies for analyzing the reliability and resilience of transportation networks. This section reviews notable contributions that apply GNN-based techniques to address critical challenges such as traffic flow prediction, traffic accident analysis, anomaly detection, criticality analysis, and decision-making and transportation management. The related work of GNN applications in transportation network in summarized in Table 2.

1) TRAFFIC FLOW PREDICTION

Predicting traffic conditions and evaluating resilience under various scenarios have been the focus of numerous studies. One prominent application in this domain is traffic flow prediction. However, most existing approaches primarily concentrate on accurate prediction, often neglecting the critical aspects of reliability and resilience in transportation networks. Only a limited number of studies have incorporated external factors, such as weather conditions, probabilistic hazard events, and traffic accidents, within the context of traffic flow prediction. This gap highlights the potential of using GNNs to provide reliability and resilience indicators by analyzing the evolution of traffic states over time.

For example, [69] introduced a spatial-temporal multi-scale GNN model, enhanced with a temporal self-attention mechanism, to predict future traffic flow. Their approach effectively captures periodic fluctuations and time-shifted dependencies in traffic data, addressing key challenges in this domain. Building on this, [72] proposed a dynamic multi-graph GNN that integrates traffic accident data into the prediction process, demonstrating the interconnectedness of traffic flow and incident patterns.

Moreover, the integration of multi-modal data further advances traffic forecasting capabilities. For instance, [68] extended the application of GNNs by combining Transformer-based models with diverse datasets, including camera feeds, satellite imagery, and weather data. This multi-modal approach enhances prediction accuracy and situational awareness. Similarly, [77] employed dynamic multi-granularity GNNs to address the non-stationarity of traffic data, particularly during extreme weather events such as heavy rainfall. These traffic forecasting frameworks provide valuable insights into road network performance under adverse conditions, improving the reliability and resilience of traffic flow prediction systems.

2) TRAFFIC ACCIDENT ANALYSIS

As discussed in Section IV-A1, traffic accident analysis is closely related to traffic flow prediction due to the frequent occurrence of accidents on transportation networks. These incidents often cause significant disruptions, leading

TABLE 2. Summary of literature on GNN applications in transportation systems.

Literature	Task	Input	Output	Problem type	Modality
[68]	traffic flow prediction	satellite imagery, weather data, and traffic data	Traffic congestion	edge regression	spatial-temporal
[69]	traffic flow prediction	historical traffic flow	future traffic flow	edge regression	spatial-temporal
[70]	traffic flow prediction	historical traffic flow, traffic accident data	future traffic flow	edge regression	spatial-temporal
[71]	traffic accident prediction	traffic history, weather data	incident occurrence indicator	edge classification	spatial-temporal
[72]	traffic accident prediction	geospatial features, intersection feature, road attributes	incident occurrence indicator	edge classification	spatial-temporal
[73]	traffic accident prediction	weather data, trip data	incident occurrence indicator	edge classification	spatial-temporal
[74]	Criticality analysis	node operational feature	link criticality ranking	edge regression	spatial
[75]	Criticality analysis	earthquake feature, bridge fragility	bridge retrofit ranking	edge regression	spatial
[76]	Criticality analysis	edge embedding, topological feature	bridge retrofit ranking	edge regression	spatial
[77]	reliability analysis	historical traffic flow, weather condition	node resilience indicator, future traffic flow	edge regression	spatial-temporal
[78]	anomaly detection	historical traffic flow	future event indicator	edge regression	spatial-temporal
[79]	anomaly detection	traffic flow, power measurements	FDIA indicator	node regression	spatial-temporal
[80]	decision-making	bridge defects and interdependencies	bridge defect label	node regression	spatial
[81]	decision-making	pavement structure features, material properties, 3D contact stress, and previous pavement responses	3D pavement response	node regression	spatial-temporal

to congestion and bottlenecks. Consequently, identifying vulnerable and accident-prone regions within transportation networks has emerged as a critical research area. By predicting areas with a high likelihood of future traffic accidents, it becomes possible to prevent and mitigate their consequences, thereby enhancing the overall functionality and resilience of transportation systems.

One of the primary challenges in traffic accident analysis lies in handling the heterogeneous data involved in accident prediction. Recent advancements in GNNs have facilitated substantial progress in this domain. For instance, DSTGCN [82] utilizes large-scale real-world data, including accident records, vehicle speeds, and point-of-interest distributions, to enhance the accuracy of traffic accident prediction. The model provides a framework for accident prediction considering both spatial and temporal evolution.

Another challenge in traffic accident analysis is the issue of sparse and incomplete data in traffic accident records. Building on the foundation of [71] and [82] introduces a multi-structured GNN model that integrates multi-modal data and employs a clustering-based data imputation method to address sparsity issues in datasets, significantly improving predictive performance.

Expanding the focus to spatial and temporal dependencies, [73] incorporated physics-based dynamics into traffic evolution modeling. Specifically, the study proposed a traffic accident prediction framework that leverages Voronoi diagrams to account for geographical spatial distributions. This approach effectively handles complex crash data structures and provides accurate predictions of crash probabilities. By enabling proactive traffic accident predictions, such frameworks support adaptive management strategies, enhancing the ability of transportation systems to withstand and recover from disruptions.

3) ANOMALY DETECTION

Compared to traffic accidents, anomaly detection encompasses a broader concept. While traffic accidents represent a specific area of anomalies within transportation systems, anomalies more generally include various abnormal behaviors caused by human factors, and external conditions. In addition to accidents resulting from human error or adverse weather conditions, modern intelligent transportation systems, enabled by IoT technologies and vehicle-to-infrastructure (V2I) communication, are increasingly susceptible to cybersecurity threats. These threats can target multiple components, such as road networks and electric vehicles (EVs).

For instance, traffic congestion is often considered one of the most significant anomalous scenarios in transportation networks. Moreover, as discussed in Section IV-A1, issues such as data imbalance and missing data frequently arise in anomaly detection [83], leading to challenges like overfitting or underfitting during the training phase, which can severely degrade model performance. To address these challenges, [78] introduced a topology-aware GNN designed to predict traffic conditions in event-driven congestion scenarios. Their model effectively tackles the sparsity issues inherent in event-driven traffic data by employing innovative preprocessing techniques. Similarly, [84] proposed a contrastive GNN-based learning framework for traffic anomaly analysis. This framework alleviates the impact of imbalanced datasets during training by enhancing the quality of latent representations and node embeddings, resulting in improved model performance. Additionally, [85] introduced a spatial-temporal inductive GNN framework to impute missing sensor data without requiring retraining, making it highly robust and flexible for real-time applications.

Furthermore, electrified transportation systems are particularly vulnerable to false data injection attacks (FDIAs), which manipulate operational decisions and subsequently degrade EV charging satisfaction rates. To alleviate this issue, [79] addressed these resilience challenges by developing a GNN autoencoder-based model for detecting FDIAs. This approach enhances resilience in both power and transportation networks, ensuring reliable operations.

These studies demonstrate the critical role of GNNs in anomaly detection in infrastructure systems, spanning a wide range of scenarios from cybersecurity attacks to traffic congestion. By addressing challenges such as data sparsity and imbalance, these approaches contribute to more robust and adaptive transportation systems.

4) CRITICALITY ANALYSIS

Criticality analysis represents another key area where GNNs have demonstrated significant utility. Often, network-wide failures result from cascading disruptions that originate in a small set of individual road links. Identifying these critical links and taking early preventative action is crucial for enhancing network resilience. However, most existing approaches rely on manually designed or heuristic metrics, such as travel time, capacity, and annual average daily traffic (AADT). While these metrics are valuable, they often fail to capture the complex interdependencies within road networks.

To address these limitations, [74] applied a random road graph model to evaluate network robustness under link failures. This approach integrates realistic road network features, such as varying link costs and structural patterns, to provide a more comprehensive understanding of network behavior. Building on this, [86] proposed a method to rank the importance of road links by modeling spatial-temporal dependencies among them. By introducing a traffic influence propagation model, they quantified each link's impact on the entire network, offering a data-driven approach to criticality assessment.

Similarly, [76] leverages edge adjacency matrices and specialized aggregation schemes to estimate edge betweenness centrality and rank critical road segments in transportation networks. Furthermore, it addresses dynamic scenarios such as disruptions and recovery sequences, making it particularly well-suited for adaptive transportation management.

These advancements highlight the potential of GNNs to improve criticality analysis by moving beyond heuristic approaches and incorporating spatial-temporal dynamics and data-driven insights.

5) DECISION-MAKING & MANAGEMENT

GNNs have also been applied to enhance decision-making and maintenance strategies in transportation systems. These decision-making methodologies generally follow two main approaches. The first integrates GNN surrogate models with meta-heuristic optimization frameworks for optimization and decision-making. For example, [75] optimized seismic retrofitting of bridges by integrating GNN surrogates with transportation equity considerations. This approach enables the prioritization of bridge retrofits based on their criticality and accessibility impacts while accounting for diverse income groups and equitable resource allocation. Similarly, [80] synthesized domain-specific ontologies with heterogeneous GNNs to predict bridge preservation activities. By leveraging structured domain knowledge, this method encapsulates implicit decision rules, offering enhanced strategies for maintaining bridge infrastructure. In addition, [81] proposed a GNN-based framework for modeling 3D asphalt concrete pavement responses (displacement, stress, and strain). This approach achieved satisfactory accuracy while significantly reducing computational time compared to the finite element methods (FEM), helping transportation agencies to accelerate their decision-making process.

The second approach involves integrating GNNs with reinforcement learning (RL) to optimize decision-making strategies, as demonstrated by traffic signal control algorithms [87], [88], [89]. These studies show that combining GNNs with RL can effectively adapt traffic signal phase timings to minimize congestion and improve flow. Furthermore, GNN-RL methods have also been applied to resource allocation problems, such as optimizing multi-agent systems in dynamic and resource-constrained environments [90]. Additionally, [91] introduced a graph-based method to improve safety and efficiency in vehicular ad-hoc networks, supporting decision-making processes for accident prevention and traffic safety.

These works highlight the potential of GNNs in addressing reliability and resilience challenges within transportation networks. They underscore the increasing importance of integrating advanced GNN techniques to enhance the robustness and adaptability of transportation systems in diverse and dynamic conditions.

B. WATER DISTRIBUTION NETWORKS

For water distribution systems (WDNs), reliability analysis ensures consistent access to clean water, particularly during extreme weather events or natural disasters. Resilience evaluation helps identify critical nodes, such as reservoirs or pumping stations, that must be protected or reinforced to prevent system-wide failures. Applications include optimizing flow dynamics, managing demand, and planning for disruptions in supply to ensure the continuity of water services under adverse conditions.

GNNs have also seen significant application in water distribution networks, the design, operation, and management of water distribution networks were originally often heavily dependent on hydraulic models. Facing the general reality of the lack of high-precision hydraulic models in water utilities, GNNs have emerged as a promising alternative to improve accuracy and efficiency. Focusing on tasks such as burst detection, leakage risk assessment, state estimation, and water quality prediction. The summary of related works of GNN applications in WDNs is shown in Table 3.

1) PIPELINE STATE ESTIMATION

Accurate state estimation is crucial for the efficient management of WDNs, particularly for analyzing water states and calculating pipeline head losses. Recent advancements in GNNs have significantly benefited state estimation in WDNs.

Both supervised and semi-supervised learning approaches have been widely employed to estimate flow rates and pressures within these networks [95], [97]. Furthermore, GNNs are incorporated with physics-informed constraints, as demonstrated by [96], simulating the functionality of traditional hydraulic simulators, providing efficient and accurate estimates of pressures and flow rates. Similarly, [102] integrated physics-guided mechanisms to enforce hydraulic constraints, resulting in improved prediction accuracy and interpretability compared to fully connected neural networks.

Similar to transportation networks, one key challenge in state estimation involves missing data, such as unavailable pressure values at certain sensor nodes. Addressing this,

TABLE 3. Summary of literature on GNN applications in water distribution networks.

Literature	Task	Input	Output	Problem type	Modality
[92]	leakage detection	nodal pressure + flow meters	binary abnormality indicator	node classification	spatial-temporal
[93]	leakage detection	partial observed nodal pressure	Reconstructed nodal pressures	node regression	spatial-temporal
[94]	leakage detection	pipeline features	predicted leakage probability	node regression	spatial-temporal
[95]	hydraulic prediction	elevation and nodal demands	flow rates, pressure	node regression	spatial
[96]	hydraulic prediction	pressure and nodal demands	nodal pressure	node regression	spatial
[97]	hydraulic prediction	nodal demand + pipe characteristics	nodal heads and pipe flows	node regression	spatial
[98]	data imputation	partial observed nodal pressure	reconstructed nodal pressure	node regression	spatial
[99]	data imputation	partial observed nodal pressure	Estimated pressure values	node regression	spatial-temporal
[59]	water quality prediction	chlorine concentrations, junction demands	chlorine concentrations	node regression	spatial-temporal
[100]	water quality prediction	water quality data	contamination probability	node classification	spatial-temporal
[101]	water quality prediction	historical water quality data	future water quality values	node regression	spatial-temporal
[102]	hydraulic prediction	historical hydraulic states	Future hydraulic states	node regression	spatial-temporal

[98] and [99] developed robust GNN models capable of handling sparse features and long-range dependencies in graphs. These advancements have made significant strides in reconstructing missing node values, enhancing the robustness of state estimation frameworks.

2) ANOMALY DETECTION

Early warning of abnormal events in water pipes like burst and leakage is beneficial to the safety and reliability of WDNs. However, challenges exist in model calibration with limited monitoring data, unseen underground conditions, or high computing requirements. GNNs have been employed as node classification models to identify abnormal events in WDN like burst and leakage detection.

For instance, [92] and [93] proposed a novel approach using spatial-temporal GNNs with features derived from measurement points like pressure and flow meters, which considers anomaly detection as binary classification tasks. Similarly, [94], [103], [104] introduced a method that incorporates pipeline features such as age, material, length, prior failures, and operating pressure to predict leakage probabilities.

Furthermore, [105] proposes a ST-GCN model, which combines large-scale network partition, optimized sensor placement, and network partition model to capture both spatial and temporal dependencies in the WDNs. This model enhances detection accuracy and significantly reduces false alarms.

3) WATER QUALITY ESTIMATION

Monitoring water quality is essential for the effective management of WDNs since it is closely related to the safety of the public. Ensuring the safety and reliability of drinking water supply requires accurate estimation of water quality across whole WDNs. However, installing water quality sensors at every junction is impractical due to their high costs [59]. Existing hydraulic model-based approaches for system state prediction face significant challenges when dealing with limited sensor data and high computational demands. Furthermore, they lack the capacity to predict the water quality at unmonitored sites or locations not included in model training.

By incorporating gated architectures and explicit hydraulic flow directionality, GNN-based models achieve higher accuracy even with sparse sensor coverage. For instance, [59] and [101] demonstrated the resilience of GNN-based approaches to sensor measurement uncertainties, showcasing their ability to generalize to unmonitored nodes and significantly expanding their applicability. Building on this, [100] utilized spatiotemporal water quality data and flow directionality within WDNs to efficiently locate contamination sources, even in scenarios with limited sensor networks.

Furthermore, in order to consider the interaction between water quality variables and environmental factors, [106] proposed integrating GNNs with reinforcement learning. Specifically, it utilizes a deep auto-encoder clustering method in the GNN model for water quality prediction. This approach not only incorporates environmental dynamics but also enhances prediction accuracy and adaptability.

These diverse applications underscore the potential of GNNs in water quality estimation. By enabling real-time monitoring, predictive maintenance, GNN-based methods enable more resilient and adaptive water quality management strategies.

4) SENSOR PLACEMENT OPTIMIZATION

Optimizing the placement of pressure sensors is critical for ensuring the safety and operational efficiency of water supply networks. Proper sensor placement enables effective monitoring, early detection of anomalies, and informed decision-making, thereby enhancing the resilience of water supply systems. Most existing studies focus on the sensitivity of monitoring systems under specific operating conditions of WSNs. However, key factors such as node connectivity and long-term hydraulic fluctuations have not been fully investigated.

In order to address these limitations, [107] proposed to select the most sensitive nodes as sensor locations within each monitoring partition of WDNs. Specifically, it incorporates node burst probability to prioritize sensor placement. Furthermore, [108] formulated the sensor placement problem as a combinatorial optimization task modeled as a quadratic unconstrained binary optimization problem. This approach leverages GNNs to solve the problem effectively, providing a novel perspective on sensor placement optimization.

The methodologies proposed in these studies offer significant benefits for stakeholders in WDNs. By analyzing the identification capabilities of monitoring systems under diverse operating conditions and economic constraints, these approaches provide valuable guidance for establishing effective and cost-efficient monitoring frameworks.

C. POWER DISTRIBUTION NETWORK

As a critical component in the infrastructure system, the power distribution networks (PDNs) serve as the bridge between power generation and end consumers. It ensures the reliable delivery of electricity to homes, businesses, industries, and public services. However, extreme weather events like snow storms, hurricanes, and heatwaves have increasingly caused significant socio-economic impacts on power grids. These events pose severe challenges to the operation and supply of electricity, particularly for critical sectors such as hospitals and transportation systems. These risks underscore the urgent need to enhance grid resilience against adverse scenarios and extreme weather events.

GNNs also have been applied to various tasks and challenges in PDNs, including system state estimation, operational risk assessment, anomaly detection, and grid optimization & control. Table 4 collectively highlights the versatility and potential of GNNs to model the complex dynamics of power grids. By integrating these advanced methodologies, power distribution networks can become more resilient, adaptive, and capable of meeting the demands of a rapidly changing environment.

1) SYSTEM STATE ESTIMATION

Similar to other infrastructure systems, PDNs require extensive monitoring and control to ensure reliable operation. This includes estimating critical parameters such as voltage magnitudes, phase angles, and other state variables. However, these systems face significant challenges, including a lack of observability and the high density of distribution networks. While data-driven approaches based on machine learning models offer potential solutions, they often struggle due to the sparse and noisy data.

To address these issues, state estimation can be formulated using GNNs as a node regression problem. Specifically, state measurement matrices are incorporated as input features to these models [123], [124], [125], [126]. This approach allows for more accurate modeling of the spatial and temporal dynamics inherent in distribution networks.

Furthermore, PDNs are also governed by the physics constraint. Incorporating physical laws into state estimation and applying physics-informed neural networks have been shown to improve accuracy significantly [127]. For example, [128] proposed a physics-informed graph learning approach that combines model-based dynamic state dynamics with spatial-temporal graph neural networks. Building on this foundation, [129] introduced a weakly supervised learning framework that integrates power flow equations into the GNN prediction. By enforcing consistency with the physics of distribution systems, this method ensures that the state estimation respects the underlying physical principles.

These models demonstrate robustness in scenarios involving cyberattacks and unobservable conditions caused by communication irregularities. By providing valuable insights during unexpected events, they enhance the resilience and reliability of power distribution networks, ensuring more effective monitoring and control under challenging conditions.

2) OPERATIONAL RISK ASSESSMENT

Operational risk assessment is a critical area in power systems, where GNNs have been effectively utilized to evaluate transient stability and operational risks [112], [117]. These risks include metrics such as load shedding indices, percolation thresholds, voltage stability, and cascading failure probability. A comprehensive risk assessment involves identifying these risks, followed by evaluating their impacts to develop mitigation strategies.

For instance, [109] developed a gated GNN for transient stability assessment, leveraging imbalanced data and incorporating a soft attention mechanism to enhance model interpretability. Similarly, [110] employed spatial-temporal GNNs to predict grid states over the next few hours, providing detailed assessments at the system, zone, and branch levels for load shedding and branch overloading. [113] and [114] combined topological and temporal features to identify system vulnerabilities and improve resilience metrics, such as load shedding indices and percolation thresholds.

As highlighted in Section IV-A4, incorporating physical laws significantly improves the accuracy and interpretability of risk assessment models. For example, [111] introduced a physics-informed GNN that incorporates physical laws into its loss function to enhance its predictive capabilities. Similarly, [130] developed a hybrid physics-based GNN framework that accounts for temporal characteristics, enabling the prediction of post-fault rotor angle trajectories based on pre-fault system states and trip locations. These models demonstrate their applicability in complex fault scenarios.

In addition to singular stability risk assessments, multi-stability assessments have become increasingly important given the rising uncertainties in power systems. To address these complexities, [131] proposed a GCN-based model that employs initial residual identity mapping to tackle multiple stability challenges, including rotor angle, voltage, frequency, and converter-driven stability.

Another significant challenge in risk analysis is generalization to multiple scenarios, specifically considering the heterogeneity of the power network. It is challenging to transfer the knowledge from one network to another. To alleviate this challenge, [132] modeled the risk estimation

TABLE 4. Summary of literature on GNN applications in power distribution networks.

Literature	Task	Input	Output	Problem type	Modality
[109]	reliability assessment	historical power, voltage, current	instability state	node classification	spatial-temporal
[110]	reliability assessment	admittance matrix, bus types and electrical parameters	active power output	node regression	spatial-temporal
[111]	cascading failure analysis	bus type, voltage, and zone	bus voltage.	node regression	spatial
[112]	cascading failure analysis	power injection values	failure indicator	edge regression	spatial-temporal
[113]	grid resilience analysis	electrical characteristics and transmission features	node/link criticality score	node/link classification	spatial
[114]	risk assessment	historical power data	node vulnerable score	node classification	spatial-temporal
[115]	anomoly detection	historical weather measurements	power outage indicator	node classification	spatial-temporal
[116]	anomoly detection	historical bus voltages, active power, and power flow	power outage indicator	graph classification	spatial-temporal
[117]	anomoly detection	historical voltage, power flows, line reactance	safety indicator	graph classification	spatial
[118]	cyber attack detection	historical power measurements	abnormality probability	node classification	spatial
[119]	cyber attack detection	communication traffic throughput	abnormality probability	node classification	spatial-temporal
[120]	cyber attack detection	historical voltage, currents	abnormality probability	node classification	spatial-temporal
[121]	cyber attack detection	power injections	abnormality probability	node classification	spatial-temporal
[122]	transient dynamics prediction	historical voltage and frequency	power grid transient trajectories	node regression	spatial-temporal

problem as a regression problem defined on a bipartite meta graph supported by a graph neural network and predicted penalties induced by outages. This approach accurately modeled penalties across a wide range of existing network topologies, demonstrating robust generalization.

These contributions highlight the efficiency of GNNs in estimating and mitigating operational risk, thereby enhancing the resilience of power distribution networks under diverse operational conditions.

3) ANOMALY DETECTION

Anomaly detection plays a critical role in maintaining the reliability and stability of PDNs. These anomalies often exhibit both spatial and temporal characteristics and can arise from various scenarios, including measurement errors, sudden load changes, and, abnormal power flows. Such anomalies can disrupt essential services and lead to significant economic losses.

To address these challenges, [133] proposed a graph learning framework designed to identify and locate power and cyber abnormal events in PDNs. Similarly, [60] introduced a spatial-temporal recurrent graph neural network that extracts features from voltage measurement data at critical buses to conduct fault event detection effectively. Furthermore, higher-order topological neural networks also have emerged as powerful tools for anomaly detection. For instance, [116] utilized Hodge-Laplacian analytics to enhance multi-node substructure representation.

Cyber-attack detection in smart grids has also seen considerable advancements with GNN methodologies. For example, [118] and [121] employed GNNs to identify stealth FDIAs and other anomalies. Extending these techniques, [120] introduced dynamic graph construction methods that improved interpretability and detection precision, which is specifically applied on grid-tied photovoltaic systems.

These applications illustrate the broad utility of GNNs in PDNs, enabling advancements in detecting outlier and predictive maintenance.

4) GRID OPTIMIZATION & CONTROL

Modern power distribution networks are transitioning from independent and passive networks to flexible and interactive systems, as highlighted by [134]. Furthermore, Server weather events pose increased challenges for maintaining the resilience and reliability of modern power systems [135]. These shifts and challenges require the system providing enhanced control and optimization based on both current and future states. In response to power outages, distribution system restoration (DSR) is triggered to quickly restore affected loads by leveraging advanced control mechanisms once outages are isolated. However, existing solutions to the DSR problem predominantly rely on detailed or approximated physical power system models [136], [137]. [138] developed a model-free GNN-RL framework to learn effective control policies for DSR problems sequentially, without requiring prior knowledge of power system parameters.

As discussed in Section IV-A5, RL has also been extensively applied within policy learning frameworks for power management, offering model-free alternatives. [139] integrated multi-agent RL with physics-informed neural networks to control voltage and learn optimal reactive power generation strategies for PV inverters. Similarly, [140] proposed the balance deep Q-network algorithm to mitigate state and action imbalances in RL.

Despite these advancements, many existing models rely heavily on architectural components tied to specific global structures, which limits their adaptability and generalization capability to variations in topologies. To address this limitation, [141] proposed a GAT-based DRL scheme to solve the multi-objective optimization dispatch problem, incorporating knowledge transfer learning across different topologies. Additionally, [142] introduced a fully decentralized GNN architecture, implementing an end-to-end data-driven approach to balance power flows, demonstrating significant flexibility and scalability.

These advancements underscore the potential of RL and GNN-based approaches to revolutionize power network management. By addressing challenges such as topology variations, state imbalances, and action coupling, these methodologies enhance the adaptability and resilience of modern power distribution networks.

D. COMMUNICATION SYSTEM

GNNs also have emerged as powerful tools in the design, operation, and management of modern communication

TABLE 5. Summary of literature on GNN applications in communication systems.

Literature	Task	Input	Output	Problem type	Modality
[143]	optimization/node classification	OD demand, competing flow	optimal routing decision	routing	spatial-temporal
[144]	optimization/edge regression	network state matrix	optimal routing decision	routing	spatial
[145]	optimization/edge regression	link bandwidth, and packet loss rate, subflow configurations	optimal routing decision	routing	spatial-temporal
[146]	optimization/node regression	channel state information	optimal power allocation strategy	Power Control	spatial-temporal
[147]	optimization	channel state information	optimal power allocation strategy	Power Control	spatial-temporal
[148]	optimization	power limit, channel state information	optimal resource allocation strategy	Power Control	spatial-temporal
[149]	edge regression	channel state information	transmitter power allocation	Power Control	spatial
[150]	edge regression	channel gain information	transmitter power allocation	Power Control	spatial-temporal
[151]	Node classification	channel state information	D2D pair activity indicator	Link Scheduling	spatial
[152]	node regression	node type and traffic parameter	throughput and latency	Network Modeling	spatial

systems. As these systems become increasingly complex and interconnected, traditional methods of analysis and optimization often struggle to address their dynamic nature and scalability requirements. By leveraging the graph-based structure of communication networks, GNNs enable more accurate, efficient, and adaptable management strategies. The summary of related GNN applications is listed in Table 5

1) POWER CONTROL

Effective resource allocation and power control are critical for optimizing performance in communication networks. However, resource allocation problems, such as power control, are inherently non-convex and computationally challenging. These problems must often be solved in real-time to adapt to the time-varying nature of wireless channels. To address these challenges, significant efforts have been dedicated to developing effective GNN-based models for wireless resource allocation and power control.

For instance, [148], [149] introduced a novel aggregation mechanism that respects the permutation invariance of interference channels. This approach requires significantly fewer training samples than traditional methods, making it highly efficient and practical for real-world applications. Similarly, [147] and [150] incorporated permutation equivariance properties into the neural architecture. This innovation enables scalability and generalization across networks of varying sizes while ensuring robustness to feature corruption and other uncertainties.

Building on these advancements, [146] proposed a parameter-sharing scheme to explicitly encode the properties of the power control policy. By leveraging this scheme, their approach enhances computational efficiency and facilitates learning policies that adapt effectively to dynamic wireless environments.

2) LINK SCHEDULING

In order to determining when and how communication links in a network should be activated, we need to first folve the link scheduling task, which is a critical resource allocation problem in communication networks. Link scheduling in communication networks is a critical optimization task that involves addressing non-convex combinatorial problems. These problems are particularly challenging as they are known to be NP-hard.

To tackle this complexity, GNNs have been effectively employed to optimize link scheduling. For instance, [151] proposed a novel GNN-RL framework within a non-differentiable pipeline. This approach was applied to link scheduling in wireless networks, demonstrating its potential to handle the computational intricacies of such tasks. [132] introduced a graph embedding-based method designed to derive near-optimal scheduling strategies with minimal computational overhead. This approach constructs fully connected directed graphs and employs both supervised and unsupervised learning techniques to achieve competitive performance.

3) ROUTING OPTIMIZATION

Similar to transportation systems, routing optimization is a critical research problem in communication networks. Efficient routing ensures optimal resource utilization and enhanced system performance. Recently, the integration of RL and GNNs have emerged as a promising approach to tackle routing optimization effectively.

For instance, [145] proposed a GNN model that integrates subflow relationships, and TCP connection features to predict throughput accurately. It provides an end-to-end solution for multipath routing optimization, ensuring seamless adaptability to dynamic network conditions. Building on this, [144] developed a framework that integrates GNNs for generalization across dynamic network topologies. By incorporating a deep RL framework, their method effectively adapts to changing network states, enabling efficient decision-making in real time. Further advancing the field, [143] introduced two complementary components: a single-agent GCN model and a multi-agent Deep Q-Network model. The first component leverages the structural information of graphs to generalize across unseen network topologies without retraining. In the meanwhile, the second component supports distributed decision-making, leading to faster convergence and improved scalability in larger networks.

4) NETWORK PERFORMANCE MODELING

GNNs have also been applied to model and predict network performance metrics such as throughput and latency. Reference [152] proposed a graph-based deep learning framework that uses graph representations of network topologies to predict TCP throughput and UDP latencies. This model generalizes across different protocols and topologies without requiring high-level protocol-specific features, making it highly versatile.

E. INTERDEPENDENT INFRASTRUCTURE SYSTEMS

As discussed in Section I, modern infrastructure systems are increasingly interconnected, with complex dependencies spanning transportation, power, communication, and water networks. While this interdependence enhances efficiency and coordination, it also introduces significant vulnerabilities. Failures in one infrastructure can trigger cascading impacts across other interconnected systems, amplifying disruptions. These interdependencies make infrastructures more susceptible to various hazards, which can lead to catastrophic consequences for communities or even entire cities.

Under the context of cyber-physical systems and the Internet of Things (IoT), seamless interaction between networks is essential for the functionality for interdependent infrastructure systems. smart water meters and pressure sensors collect real-time data on flow rates and water quality, transmitting this information to data centers via wireless communication systems powered by PDNs. Similarly, sensors embedded in vehicles and road infrastructure, such as smart traffic signals, road cameras, and GPS devices, gather critical data on traffic flow, which is transmitted through communication systems.

The growing reliance on interconnected infrastructure underscores the importance of understanding and addressing their vulnerabilities. Identifying weak points and interdependencies within these networks is crucial for mitigating risks and enhancing resilience. By leveraging GNNs, researchers and policymakers can gain deeper insights into the dynamics of interconnected systems, enabling proactive measures to address potential disruptions. A summary of related work is shown in Table 6. The main objectives of GNN applications in interdependent infrastructure systems are twofold: avoiding cascading failures and detecting anomaly events.

1) CASCADING FAILURE MITIGATION

Understanding and mitigating cascading failures in interdependent systems is essential for enhancing the resilience of modern infrastructure. [155] addressed this challenge by developing a suite of six resilience metrics to quantify interdependencies between PDSs and WDSs. Their study evaluates cascading impacts by accounting for spatial configurations, distributed power sources, and water storage capacity. To further improve disaster resilience, [90] integrated heterogeneous GNNs with multi-agent RL to optimize repair policies for coupled power-water-transportation systems. Building on these insights, [156], [157] explored the impact of limiting interdependent edges in communication and power networks. These frameworks underscore the critical role of interdependence during system degradation and recovery phases, offering valuable insights into system vulnerabilities and potential intervention strategies.

These contributions collectively demonstrate the potential of GNN-based approaches in resilience analytics. By quantifying interdependencies, optimizing repair policies, and mitigating cascading failures, these frameworks pave the way for more robust and adaptive infrastructure systems.

2) ANOMALY DETECTION

Anomaly detection is also a common task in interdependent infrastructure systems. As discussed in Section IV-C3, smart power grids and electrified transportation systems are particularly vulnerable to FDIAs, which can cause threats to intelligent transportation systems. To mitigate these threats, [79] and [158] developed GNN autoencoder-based models for FDIA detection. These approaches enhance the resilience of power and transportation networks, ensuring reliable operations even in the face of sophisticated cyber threats. Reference [79] integrated graph autoencoders with Chebyshev graph convolutional layers to detect FDIAs in coupled power-transportation networks.

Similarly, [83] utilized spatio-temporal GNNs to detect anomalies in coupled power-transportation system, effectively addressing challenges such as dynamic interconnections and heterogeneous data. In another study, [78] proposed a topology-aware GNN for predicting event-driven traffic conditions in smart power grids, enabling proactive resource management and timely interventions. Reference [153] introduced a GNN-based system that models interdependent networks as heterogeneous graphs, which enables targeted and effective interventions for transportation systems.

These studies collectively demonstrate the power of GNNs in detecting anomalies within multi-scale interdependent infrastructure system. By leveraging spatio-temporal features and heterogeneous data, these models offer innovative solutions to maintain the resilience and reliability of interdependent systems.

V. FUTURE DIRECTIONS

The application of graph neural networks in infrastructure systems has shown remarkable promise in addressing challenges related to resilience, optimization, and anomaly detection. However, several critical areas warrant further exploration to fully realize the potential of GNNs in practical deployment. Below, we discuss key future directions based on current challenges and emerging needs:

A. LEVERAGING SMALL DATASETS

One of the primary challenges in applying GNNs to infrastructure systems is the limited availability of high-resolution and labeled datasets. While many infrastructure systems, such as transportation networks and power grids, generate vast amounts of raw data, these datasets often lack meaningful labels or are recorded at coarse resolutions that limit their direct applicability for advanced GNN training. Self-supervised and semi-supervised learning approaches can unlock the potential of these unlabeled datasets by creating pretext tasks that exploit structural and temporal dependencies inherent in infrastructure networks. For instance, contrastive learning on network states or temporal prediction tasks can

TABLE 6. Summary of literature on GNN applications in interdependent infrastructure systems.

Literature	Task	Interdendency	Input	Output	Problem type	Modality
[153]	node classification	transportation + power	topological features, traffic data	node vulnerability score	Criticality analysis	spatial
[154]	graph generation	supply chain + power	supply/demand information, link capacity and cost	graph resilience metric	graph generation	spatial
[155]	node regression	power + water	system configuration, water storage capacity, power sources	node resilience metrics	resilience analysis	spatial-temporal
[83]	node/edge classification	transportation+ power	assets locations, traffic flow, energy consumption	node anomaly scores	anomoly detection	spatial-temporal
[79]	Node classification	transportation + power	power measurements, traffic volumes	FDIA Detection probability	anomoly detection	spatial-temporal
[78]	node regression	transportation + power	historical event-driven traffic flow	congestion indicator	anomoly detection	spatial-temporal
[90]	sequential-decision making	power + water + transportation	damage states, restoration parameters	Optimal repair sequence	decision making	spatial-temporal
[156]	graph regression	commuication + power	topological feature, attack parameters	node failure ratio, risk mitigation strategies.	resilience analysis	spatial-temporal
[157]	edge classification	communication + power	topological feature, disaster data	link interdependency probability	network optimization	spatial-temporal

enable GNNs to learn more robust representations without requiring costly annotation.

Beyond semi-supervised methods, transfer learning and domain adaptation represent promising directions for alleviating the scarcity of labeled data in specific regions or infrastructure domains. A GNN trained on well-studied urban transportation data could, for example, be adapted to smaller cities with limited data availability. Similarly, simulation-driven data augmentation and synthetic graph generation can provide rich training samples that mimic rare but critical scenarios such as system failures, extreme weather events, or cascading disruptions. By combining real-world datasets with simulation outputs and domain knowledge, it is possible to construct hybrid training corpora that significantly expand the effective sample size available for GNN development.

B. IMPROVING COMPUTATIONAL SCALABILITY

As infrastructure systems grow in scale and complexity, the computational requirements of GNNs become increasingly burdensome. For instance, regional power grids or national highway systems can consist of millions of nodes and edges, making naive applications of GNNs infeasible. To address this, scalable graph processing methods such as sampling-based training, graph sparsification, and hierarchical pooling should be further explored. These methods allow models to focus on the most structurally significant components of the graph without losing critical system information. Additionally, distributed and parallel GNN frameworks can help leverage cloud and edge computing resources, ensuring efficient handling of massive datasets across decentralized environments.

Another critical direction is the development of GNNs capable of online and streaming graph learning. Infrastructure systems are dynamic and continuously evolving, with new data arriving in real time from sensors, IoT devices, and monitoring systems. Traditional batch training pipelines struggle to adapt quickly to these dynamic conditions. Online learning paradigms that update models incrementally as new data streams in will be crucial for enabling timely insights.

C. ENHANCING GNN INTERPRETABILITY

Interpretability remains a central challenge for GNN adoption in critical infrastructure systems. Decision-makers such as engineers, city planners, and regulators require more than just accurate predictions; they need transparent reasoning that explains why certain system states or interventions are prioritized. Techniques such as attention mechanisms and saliency maps offer a promising first step by highlighting the most influential nodes, edges, or temporal sequences in the model's reasoning process. By providing explanations that align with domain expertise, GNNs can improve trust and acceptance among practitioners tasked with deploying these models in safety-critical applications.

In addition, emerging techniques for causal interpretability and counterfactual reasoning can be applied to infrastructure settings. Such insights not only provide interpretability but also enhance the utility of GNNs for planning and risk management. Furthermore, an interpretable surrogate model may serve as an accessible decision-support tools that balance predictive accuracy with clarity.

D. INTEGRATING MULTI-MODAL DATA

Infrastructure systems operate in environments shaped by multiple interdependent factors, ranging from physical connectivity and sensor readings to socioeconomic and environmental conditions. While traditional GNN applications often focus solely on graph-structured data, significant improvements can be achieved by integrating heterogeneous data modalities. For example, satellite imagery provides spatial context, IoT devices capture real-time operational data, and climate models supply weather-related risk information. The fusion of these complementary data streams with graph structures can yield richer and more holistic system representations.

Designing architectures that can effectively handle this integration remains a key research frontier. Graph-based fusion mechanisms, multimodal attention layers, and shared latent representation spaces are promising avenues for combining diverse information sources. A transportation GNN that simultaneously processes road connectivity, GPS trajectories, and weather conditions, for instance, can more accurately predict congestion patterns or vulnerability under extreme events. Cross-modal alignment techniques will be essential for resolving discrepancies in resolution, scale, and noise across modalities.

E. MODELING DYNAMICS AND MULTI-HAZARD SCENARIOS

Infrastructure systems are not static, and they evolve over time and are subject to continuous changes due to demand fluctuations, maintenance activities, and unexpected disturbances. Traditional GNNs often treat these systems as static or rely on short time horizons, which limits their ability to capture long-term temporal dependencies. Future research should emphasize spatio-temporal GNNs capable of modeling complex dynamics at multiple timescales. This includes methods that can adaptively update embeddings as new data arrives, providing real-time situational awareness in evolving contexts such as traffic management or power grid stability.

Furthermore, natural disasters like earthquakes and floods often occur in conjunction with secondary risks such as infrastructure failures or cyber-attacks, amplifying their impact. GNN frameworks that account for such multi-hazard interactions will provide more realistic resilience assessments and preparedness strategies. Integrating probabilistic modeling and uncertainty quantification with GNN architectures can further help capture the variability and unpredictability of hazard interactions, guiding robust and adaptive infrastructure planning.

F. HUMAN-IN-THE-LOOP AND DECISION SUPPORT

Infrastructure management decisions are rarely made by algorithms alone; they involve close collaboration between human experts and decision-support systems. A promising direction is to design GNN frameworks that incorporate human expertise during both the training and deployment phases. Active learning approaches, where human feedback is used to label or validate uncertain predictions, can significantly improve performance in data-scarce environments. Similarly, visualization tools that translate GNN outputs into human-understandable insights can empower operators to explore alternative scenarios and validate model recommendations.

Beyond technical assistance, GNN-based tools should be aligned with policy and governance processes. Decision-makers often need to evaluate trade-offs across economic, environmental, and social objectives, which go beyond pure technical optimization. Embedding GNNs into interactive platforms for policy evaluation, scenario testing, and participatory planning can help bridge the gap between algorithmic predictions and real-world actions. Such human-in-the-loop approaches will not only increase trust and usability but also ensure that AI-driven infrastructure decisions are aligned with societal values and regulatory requirements.

G. ROBUSTNESS, RELIABILITY, AND SAFETY

For real-world deployment, GNN models must be robust against noisy data, distribution shifts, and adversarial attacks. Future research should investigate certified robustness guarantees, uncertainty quantification, and calibration methods to ensure reliable performance under diverse operating conditions. Incorporating physics-based constraints and domain knowledge into GNN architectures may improve safety by enforcing physically plausible outputs. Developing standards and benchmarks for robustness in infrastructure-related GNN tasks will also be important for evaluating and comparing methods in practice.

H. INTEGRATION WITH DIGITAL TWINS

Digital twins are becoming increasingly central to infrastructure management. GNNs, with their ability to model connectivity and dynamic interactions, are well suited to serve as the computational backbone of such digital twins. By capturing both static topology and dynamic flows, GNN-powered digital twins could enable predictive maintenance, anomaly detection, and scenario simulation. This would provide decision-makers with actionable foresight, allowing them to anticipate problems before they occur.

Future research should focus on the seamless integration of GNN models with digital twin platforms at scale. This includes challenges in synchronizing data streams, handling latency, and ensuring model adaptivity in response to new information. Coupling GNN-driven digital twins with optimization and control frameworks could further support real-time interventions, such as dynamic traffic routing or grid load balancing. By bridging the physical and virtual worlds, GNN-enhanced digital twins hold the potential to revolutionize infrastructure monitoring and management.

I. TOWARDS SUSTAINABLE AND EQUITABLE INFRASTRUCTURE

Infrastructure systems play a critical role in shaping the sustainability and equity of urban and regional development. Future GNN research should therefore extend beyond efficiency and resilience to explicitly incorporate environmental and social objectives. For instance, GNNs could be employed to design transportation policies that minimize carbon emissions or optimize energy distribution in ways that prioritize renewable integration. By embedding sustainability metrics into the optimization process, GNNs can contribute directly to climate mitigation and adaptation goals.

At the same time, equity considerations are increasingly important in infrastructure planning, as marginalized communities often bear disproportionate burdens from service disruptions or poor accessibility. GNN frameworks that incorporate demographic and socioeconomic attributes can help identify inequities and support more inclusive planning. Fairness-aware training objectives and constraints can ensure that model recommendations distribute benefits equitably across different populations. Such approaches will align GNN research with the broader agenda of creating sustainable, resilient, and just infrastructure systems for the future.

VI. DISCUSSION AND CONCLUSION

Graph neural networks have emerged as a powerful tool in modern infrastructure systems, enabling more robust, adaptive, and efficient solutions to critical challenges across transportation, water distribution, power distribution, communication, and interdependent infrastructure networks. By leveraging the inherent graph-based structures of these systems, GNNs have proven to be highly effective in addressing complex problems such as cascading failure mitigation, anomaly detection, criticality analysis, and optimization of resource allocation and control strategies.

In transportation networks, GNNs have advanced the fields of traffic flow prediction, accident analysis, and anomaly detection, offering enhanced predictive accuracy and resilience by incorporating spatial-temporal dynamics, multimodal data, and real-time adaptability. Similarly, in water distribution networks, GNNs have demonstrated remarkable capabilities in pipeline state estimation, burst and leakage detection, water quality estimation, and sensor placement optimization, addressing challenges posed by sparse data, high computational demands, and system complexity.

In power distribution systems, GNNs have been instrumental in system state estimation, operational risk assessment, anomaly detection, and grid optimization. By integrating physics-informed constraints and reinforcement learning frameworks, GNNs have significantly enhanced the reliability, scalability, and adaptability of power grids under dynamic and adverse conditions. Communication systems have also benefited from GNN-based advancements in power control, link scheduling, routing optimization, and performance modeling, offering efficient solutions to traditionally intractable problems such as NP-hard combinatorial tasks.

The growing interdependence of infrastructure systems, enabled by cyber-physical systems and the Internet of Things (IoT), has introduced new vulnerabilities while also amplifying the need for advanced analytical tools. GNNs have risen to this challenge by enabling the analysis of cascading failures, detection of anomalies, and optimization of repair and restoration strategies in interdependent systems. Through spatio-temporal modeling, multi-modal integration, and resilience-focused frameworks, GNNs provide the necessary insights to mitigate risks and enhance the robustness of coupled systems.

Across all infrastructure domains, the reviewed studies collectively highlight the transformative potential of GNNs in tackling the challenges posed by increasing complexity, interdependencies, and dynamic conditions. By bridging the gap between traditional heuristic approaches and advanced data-driven methods, GNNs are poised to revolutionize infrastructure management, offering innovative solutions for real-time monitoring, predictive maintenance, and decision-making. The future of infrastructure systems lies in harnessing the full potential of GNNs to build safer, smarter, and more resilient communities.

REFERENCES

[1] Y. Popova, "Relations between wellbeing and transport infrastructure of the country," *Proc. Eng.*, vol. 178, pp. 579–588, 2017, doi: 10.1016/j.proeng.2017.01.112.

[2] S. Daulat, M. M. Rokstad, A. Klein-Paste, J. Langeveld, and F. Tscheikner-Gratl, "Challenges of integrated multi-infrastructure asset management: A review of pavement, sewer, and water distribution networks," *Struct. Infrastruct. Eng.*, vol. 20, no. 4, pp. 546–565, Apr. 2024, doi: 10.1080/15732479.2022.2119480.

[3] L. Zhao, J. Li, T. Zhang, P. Feng, X. Wang, and J. Shen, "Multi-method combined analysis of urban flood risks and its influencing factors under low impact development," *J. Hydrol.*, vol. 632, Mar. 2024, Art. no. 130927, doi: 10.1016/j.jhydrol.2024.130927.

[4] F. Liu, W. Ding, Y. Qiao, and L. Wang, "Transfer learning-based encoder–decoder model with visual explanations for infrastructure crack segmentation: New open database and comprehensive evaluation," *Underground Space*, vol. 17, pp. 60–81, Aug. 2024, doi: 10.1016/j.undsp.2023.09.012.

[5] M. Kiani-Moghaddam, M. Shivaie, and A. Arabkoohsar, "Towards a new effective strategy to obtain optimal radial structure in power distribution networks: Graph theory-based topology assessment," *Int. J. Electr. Power Energy Syst.*, vol. 143, Dec. 2022, Art. no. 108484, doi: 10.1016/j.ijepes.2022.108484.

[6] Z. Lv, B. Hu, and H. Lv, "Infrastructure monitoring and operation for smart cities based on IoT system," *IEEE Trans. Ind. Informat.*, vol. 16, no. 3, pp. 1957–1962, Mar. 2020, doi: 10.1109/TII.2019.2913535.

[7] M. K. Habib and C. Chimsom, "CPS: Role, characteristics, architectures and future potentials," *Proc. Comput. Sci.*, vol. 200, pp. 1347–1358, 2022, doi: 10.1016/j.procs.2022.01.336.

[8] T. Liu and H. Meidani, "Physics-informed neural networks for system identification of structural systems with a multiphysics damping model," *J. Eng. Mech.*, vol. 149, no. 10, Oct. 2023, Art. no. 04023079.

[9] K. Gulati, R. S. Kumar Boddu, D. Kapila, S. L. Bangare, N. Chandnani, and G. Saravanan, "A review paper on wireless sensor network techniques in Internet of Things (IoT)," *Mater. Today, Proc.*, vol. 51, pp. 161–165, 2022, doi: 10.1016/j.matpr.2021.05.067.

[10] E. Rome, M. Bogen, D. Lückerath, O. Ullrich, R. Worst, E. Streberová, M. Dumonteil, M. Mendizabal, B. Abajo, E. Feliú, P. Bösch, A. Connelly, and J. Carter, "Risk-based analysis of the vulnerability of urban infrastructure to the consequences of climate change," *Crit. Infrastructure Secur. Resilience, Theories*, pp. 55–75, Jan. 2019, doi: 10.1007/978-3-030-00024-0_4.

[11] M. A. Ahmed, B. Sridharan, N. Saha, S. A. Sannasiraj, and S. N. Kuiry, "Assessment of coastal vulnerability for extreme events," *Int. J. Disaster Risk Reduction*, vol. 82, Nov. 2022, Art. no. 103341, doi: 10.1016/j.ijdrr.2022.103341.

[12] F. Liu, J. Liu, and L. Wang, "Asphalt pavement crack detection based on convolutional neural network and infrared thermography," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22145–22155, Nov. 2022, doi: 10.1109/TITS.2022.3142393.

[13] F. Liu, J. Liu, and L. Wang, "Asphalt pavement fatigue crack severity classification by infrared thermography and deep learning," *Autom. Construct.*, vol. 143, Nov. 2022, Art. no. 104575, doi: 10.1016/j.autcon.2022.104575.

[14] S. Liu, C. Yin, D. Chen, H. Lv, and Q. Zhang, "Cascading failure in multiple critical infrastructure interdependent networks of syncretic railway system," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 5740–5753, Jun. 2022, doi: 10.1109/TITS.2021.3057404.

[15] F. Liu, J. Liu, and L. Wang, "Deep learning and infrared thermography for asphalt pavement crack severity classification," *Autom. Construct.*, vol. 140, Aug. 2022, Art. no. 104383, doi: 10.1016/j.autcon.2022.104383.

[16] F. Liu, J. Liu, L. Wang, and I. L. Al-Qadi, "Multiple-type distress detection in asphalt concrete pavement using infrared thermography and deep learning," *Autom. Construct.*, vol. 161, May 2024, Art. no. 105355, doi: 10.1016/j.autcon.2024.105355.

[17] M. Haggag, M. Ezzeldin, W. El-Dakhakhni, and E. Hassini, "Resilient cities critical infrastructure interdependence: A meta-research," *Sustain. Resilient Infrastruct.*, vol. 7, no. 4, pp. 291–312, Jul. 2022, doi: 10.1080/23789689.2020.1795571.

[18] T. Liu and H. Meidani, "Graph neural networks for travel distance estimation and route recommendation under probabilistic hazards," *Int. J. Transp. Sci. Technol.*, Feb. 2025, doi: 10.1016/j.ijtst.2025.02.006.

[19] J. Luo, G. Wang, G. Li, and G. Pesce, "Transport infrastructure connectivity and conflict resolution: A machine learning analysis," *Neural Comput. Appl.*, vol. 34, no. 9, pp. 6585–6601, May 2022, doi: 10.1007/s00521-021-06015-5.

[20] Z. Xu, Q. Bai, Y. Shao, A. Hu, and Z. Dong, "A review on passenger emergency evacuation from multimodal transportation hubs," *J. Traffic Transp. Eng. English Ed.*, vol. 9, no. 4, pp. 591–607, Aug. 2022, doi: 10.1016/j.jtte.2022.02.001.

[21] S. Rahman, S. Saha, S. N. Islam, M. T. Arif, M. Mosadeghy, M. E. Haque, and A. M. T. Oo, "Analysis of power grid voltage stability with high penetration of solar PV systems," *IEEE Trans. Ind. Appl.*, vol. 57, no. 3, pp. 2245–2257, May 2021, doi: 10.1109/TIA.2021.3066326.

[22] T. Dawood, E. Elwakil, H. M. Novoa, and J. F. G. Delgado, "Toward urban sustainability and clean potable water: Prediction of water quality via artificial neural networks," *J. Cleaner Prod.*, vol. 291, Apr. 2021, Art. no. 125266, doi: 10.1016/j.jclepro.2020.125266.

[23] X. Hu, Y. Han, B. Yu, Z. Geng, and J. Fan, "Novel leakage detection and water loss management of urban water supply network using multiscale neural networks," *J. Cleaner Prod.*, vol. 278, Jan. 2021, Art. no. 123611, doi: 10.1016/j.jclepro.2020.123611.

[24] X. Fan, X. Wang, X. Zhang, and X. Yu, "Machine learning based water pipe failure prediction: The effects of engineering, geology, climate and socio-economic factors," *Rel. Eng. Syst. Saf.*, vol. 219, Mar. 2022, Art. no. 108185, doi: 10.1016/j.ress.2021.108185.

[25] M. Lehto, "Cyber-attacks against critical infrastructure," in *Proc. Cyber Security, Crit. Infrastruct. Protection*. Springer, 2022, pp. 3–42, doi: 10.1007/978-3-030-91293-2_1.

[26] A. Mottahedi, F. Sereshki, M. Ataei, A. N. Qarahasanlou, and A. Barabadi, "Resilience estimation of critical infrastructure systems: Application of expert judgment," *Rel. Eng. Syst. Saf.*, vol. 215, Nov. 2021, Art. no. 107849, doi: 10.1016/j.ress.2021.107849.

[27] W. De-Gen, "Review on the resilient city research overseas," *Prog. Geography*, pp. 1245–1255, 2012.

[28] K. E. Weick and K. M. Sutcliffe, *Managing the Unexpected: Resilient Performance in an Age of Uncertainty*. Hoboken, NJ, USA: Wiley, 2011.

[29] C. Poulin and M. B. Kane, "Infrastructure resilience curves: Performance measures and summary metrics," *Rel. Eng. Syst. Saf.*, vol. 216, Dec. 2021, Art. no. 107926, doi: 10.1016/j.ress.2021.107926.

[30] B. Frommer, "Climate change and the resilient society: Utopia or realistic option for German regions?" *Natural Hazards*, vol. 67, no. 1, pp. 99–115, 2013, doi: 10.1007/s11069-012-0421-0.

[31] C. Wan, Z. Yang, D. Zhang, X. Yan, and S. Fan, "Resilience in transportation systems: A systematic review and future directions," *Transp. Rev.*, vol. 38, no. 4, pp. 479–498, Jul. 2018, doi: 10.1080/01441647.2017.1383532.

[32] I. E. Daoui, M. Itmi, A. E. Hami, N. Hmina, and T. Mazri, "A deterministic approach for systems-of-systems resilience quantification," *Int. J. Crit. Infrastructures*, vol. 14, no. 1, pp. 80–99, 2018, doi: 10.1504/ijcis.2018.090654.

[33] M. Abbasnejadfard, M. Bastami, M. Abbasnejadfard, and S. Borzoo, "Novel deterministic and probabilistic resilience assessment measures for engineering and infrastructure systems based on the economic impacts," *Int. J. Disaster Risk Reduction*, vol. 75, Jun. 2022, Art. no. 102956, doi: 10.1016/j.ijdrr.2022.102956.

[34] A. Kumar, K. Haque, S. Mishra, and M. M. Golias, "Multi-criteria based approach to identify critical links in a transportation network," *Case Stud. Transp. Policy*, vol. 7, no. 3, pp. 519–530, Sep. 2019, doi: 10.1016/j.cstp.2019.07.006.

[35] C. J. Applegate and I. Tien, "Framework for probabilistic vulnerability analysis of interdependent infrastructure systems," *J. Comput. Civil Eng.*, vol. 33, no. 1, Jan. 2019, Art. no. 04018058, doi: 10.1061/(asce)cp.1943-5487.0000801.

[36] A. Yehia and O. Swei, "Probabilistic infrastructure performance models: An iterative-methods approach," *Transp. Res. C, Emerg. Technol.*, vol. 111, pp. 245–254, Feb. 2020, doi: 10.1016/j.trc.2019.12.019.

[37] Z. Guo, Y. Shen, M. Aloqaily, Y. Jararweh, and K. Yu, "Probabilistic inference-based modeling for sustainable environmental systems under hybrid cloud infrastructure," *Simul. Model. Pract. Theory*, vol. 107, Feb. 2021, Art. no. 102215, doi: 10.1016/j.simpat.2020.102215.

[38] A. R. Abbasi and M. Mohammadi, "Probabilistic load flow in distribution networks: An updated and comprehensive review with a new classification proposal," *Electric Power Syst. Res.*, vol. 222, Sep. 2023, Art. no. 109497, doi: 10.1016/j.epsr.2023.109497.

[39] A. Eldosouky, W. Saad, and N. Mandayam, "Resilient critical infrastructure: Bayesian network analysis and contract-based optimization," *Rel. Eng. Syst. Saf.*, vol. 205, Jan. 2021, Art. no. 107243, doi: 10.1016/j.ress.2020.107243.

[40] J.-Z. Yu, M. Whitman, A. Kermanshah, and H. Baroud, "A hierarchical Bayesian approach for assessing infrastructure networks serviceability under uncertainty: A case study of water distribution systems," *Rel. Eng. Syst. Saf.*, vol. 215, Nov. 2021, Art. no. 107735, doi: 10.1016/j.ress.2021.107735.

[41] H. Chen, L. Zhang, Q. Liu, H. Wang, and X. Dai, "Simulation-based vulnerability assessment in transit systems with cascade failures," *J. Cleaner Prod.*, vol. 295, May 2021, Art. no. 126441, doi: 10.1016/j.jclepro.2021.126441.

[42] J. Behrensdorf, T.-E. Regenhardt, M. Broggi, and M. Beer, "Numerically efficient computation of the survival signature for the reliability analysis of large networks," *Rel. Eng. Syst. Saf.*, vol. 216, Dec. 2021, Art. no. 107935, doi: 10.1016/j.ress.2021.107935.

[43] F. Di Maio, C. Pettorossi, and E. Zio, "Entropy-driven Monte Carlo simulation method for approximating the survival signature of complex infrastructures," *Rel. Eng. Syst. Saf.*, vol. 231, Mar. 2023, Art. no. 108982, doi: 10.1016/j.ress.2022.108982.

[44] D. Chen, P. Wawrzynski, and Z. Lv, "Cyber security in smart cities: A review of deep learning-based applications and case studies," *Sustain. Cities Soc.*, vol. 66, Mar. 2021, Art. no. 102655, doi: 10.1016/j.scs.2020.102655.

[45] M. A. Nabian and H. Meidani, "Deep learning for accelerated seismic reliability analysis of transportation networks," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 33, no. 6, pp. 443–458, Jun. 2018, doi: 10.1111/mice.12359.

[46] K. Dick, L. Russell, Y. Souley Dosso, F. Kwamena, and J. R. Green, "Deep learning for critical infrastructure resilience," *J. Infrastruct. Syst.*, vol. 25, no. 2, Jun. 2019, Art. no. 05019003, doi: 10.1061/(asce)is.1943-555x.0000477.

[47] F. Liu and L. Wang, "UNet-based model for crack detection integrating visual explanations," *Construction Building Mater.*, vol. 322, Mar. 2022, Art. no. 126265, doi: 10.1016/j.conbuildmat.2021.126265.

[48] Z. Zohrevand, U. Glässer, M. A. Tayebi, H. Y. Shahir, M. Shirmaleki, and A. Y. Shahir, "Deep learning based forecasting of critical infrastructure data," in *Proc. ACM Conf. Inf. Knowl. Manage.*, Nov. 2017, pp. 1129–1138, doi: 10.1145/3132847.3133031.

[49] F. Liu, J. Li, and L. Wang, "PI-LSTM: Physics-informed long short-term memory network for structural response modeling," *Eng. Struct.*, vol. 292, Oct. 2023, Art. no. 116500, doi: 10.1016/j.engstruct.2023.116500.

[50] H. Wang, J. Chen, T. Pan, Z. Dong, L. Zhang, R. Jiang, and X. Song, "Evaluating the generalization ability of spatiotemporal model in urban scenario," 2024, *arXiv:2410.04740*.

[51] G. Jin, Y. Liang, Y. Fang, Z. Shao, J. Huang, J. Zhang, and Y. Zheng, "Spatio-temporal graph neural networks for predictive learning in urban computing: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 10, pp. 5388–5408, Oct. 2024, doi: 10.1109/TKDE.2023.3333824.

[52] C. Nwankpa, W. Ijomah, A. Gachagan, and S. Marshall, "Activation functions: Comparison of trends in practice and research for deep learning," 2018, *arXiv:1811.03378*.

[53] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

[54] P. Veličkovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.

[55] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 1025–1035.

[56] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: 10.1162/neco.1997.9.8.1735.

[57] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.

[58] X. Ma, H. Zhou, and Z. Li, "On the resilience of modern power systems: A complex network perspective," *Renew. Sustain. Energy Rev.*, vol. 152, Dec. 2021, Art. no. 111646, doi: 10.1016/j.rser.2021.111646.

[59] A. K. Salem, A. F. Taha, and A. A. Abokifa, "Graph neural networks-based dynamic water quality state estimation in water distribution networks," *Eng. Appl. Artif. Intell.*, vol. 138, Dec. 2024, Art. no. 109426, doi: 10.1016/j.engappai.2024.109426.

[60] B. L. H. Nguyen, T. V. Vu, T.-T. Nguyen, M. Panwar, and R. Hovsapian, "Spatial–temporal recurrent graph neural networks for fault diagnostics (don't short) in power distribution systems," *IEEE Access*, vol. 11, pp. 46039–46050, 2023, doi: 10.1109/ACCESS.2023.3273292.

[61] F. Liu, Y. Xu, J. Li, and L. Wang, "Graph neural network–based spatiotemporal structural response modeling in buildings," *J. Comput. Civil Eng.*, vol. 39, no. 2, Mar. 2025, Art. no. 04025006, doi: 10.1061/jccee5.cpeng-6229.

[62] T. Liu and H. Meidani, "End-to-end heterogeneous graph neural networks for traffic assignment," *Transp. Res. C, Emerg. Technol.*, vol. 165, Aug. 2024, Art. no. 104695.

[63] T. Liu and H. Meidani, "Multi-class traffic assignment using multi-view heterogeneous graph attention networks," *Expert Syst. Appl.*, vol. 286, Aug. 2025, Art. no. 128072, doi: 10.1016/j.eswa.2025.128072.

[64] N. Kovačev, M. Gavrić, and I. Lendák, "Algorithm for visualizing substation areas in electric power systems," *Expert Syst. Appl.*, vol. 212, Feb. 2023, Art. no. 118733, doi: 10.1016/j.eswa.2022.118733.

[65] J. Suárez-Varela, P. Almasan, M. Ferriol-Galmés, K. Rusek, F. Geyer, X. Cheng, X. Shi, S. Xiao, F. Scarselli, A. Cabellos-Aparicio, and P. Barlet-Ros, "Graph neural networks for communication networks: Context, use cases and opportunities," *IEEE Netw.*, vol. 37, no. 3, pp. 146–153, May 2023, doi: 10.1109/MNET.123.2100773.

[66] T. Liu and H. Meidani, "Graph neural network surrogate for seismic reliability analysis of highway bridge systems," *J. Infrastruct. Syst.*, vol. 30, no. 4, Dec. 2024, Art. no. 05024004, doi: 10.1061/jitse4.iseng-2264.

[67] Y. Dong and D. M. Frangopol, "Probabilistic assessment of an interdependent healthcare–bridge network system under seismic hazard," *Struct. Infrastruct. Eng.*, vol. 13, no. 1, pp. 160–170, 2017.

[68] S. Dhanasekaran, D. Gopal, J. Logeshwaran, N. Ramya, and A. O. Salau, "Multi-model traffic forecasting in smart cities using graph neural networks and transformer-based multi-source visual fusion for intelligent transportation management," *Int. J. Intell. Transp. Syst. Res.*, vol. 22, no. 3, pp. 1–24, Dec. 2024, doi: 10.1007/s13177-024-00413-4.

[69] C. Diao, D. Zhang, W. Liang, K.-C. Li, Y. Hong, and J.-L. Gaudiot, "A novel spatial–temporal multi-scale alignment graph neural network security model for vehicles prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 904–914, Jan. 2023, doi: 10.1109/TITS.2022.3140229.

[70] Y. Ye, Y. Xiao, Y. Zhou, S. Li, Y. Zang, and Y. Zhang, "Dynamic multi-graph neural network for traffic flow prediction incorporating traffic accidents," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121101, doi: 10.1016/j.eswa.2023.121101.

[71] T. Tran, D. He, J. Kim, and M. Hickman, "MSGNN: A multi-structured graph neural network model for real-time incident prediction in large traffic networks," *Transp. Res. C, Emerg. Technol.*, vol. 156, Nov. 2023, Art. no. 104354, doi: 10.1016/j.trc.2023.104354.

[72] B. Huang, B. Hooi, and K. Shu, "TAP: A comprehensive data repository for traffic accident prediction in road networks," in *Proc. 31st ACM Int. Conf. Adv. Geographic Inf. Syst.*, 2023, pp. 1–4, doi: 10.1145/3589132.3625655.

[73] J. Gan, Q. Yang, D. Zhang, L. Li, X. Qu, and B. Ran, "A novel Voronoi-based spatio-temporal graph convolutional network for traffic crash prediction considering geographical spatial distributions," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 12, pp. 21723–21736, Dec. 2024, doi: 10.1109/TITS.2024.3452275.

[74] P. Y. R. Sohouenou, P. Christidis, A. Christodoulou, L. A. C. Neves, and D. L. Presti, "Using a random road graph model to understand road networks robustness to link failures," *Int. J. Crit. Infrastruct. Protection*, vol. 29, Jun. 2020, Art. no. 100353, doi: 10.1016/j.ijcip.2020.100353.

[75] T. Liu and H. Meidani, "Optimizing seismic retrofit of bridges: Integrating efficient graph neural network surrogates and transportation equity," in *Proc. Cyber-Phys. Syst. Internet Things Week*, 2023, pp. 367–372, doi: 10.1145/3576914.3587503.

[76] D. Jana, S. Malama, S. Narasimhan, and E. Taciroglu, "Edge-based graph neural network for ranking critical road segments in a network," *PLoS ONE*, vol. 18, no. 12, Dec. 2023, Art. no. e0296045, doi: 10.1371/journal.pone.0296045.

[77] D. Zang, Y. Ding, J. Zhao, K. Tang, and H. Zhu, "Predictive resilience assessment of road networks based on dynamic multi-granularity graph neural network," *Neurocomputing*, vol. 601, Oct. 2024, Art. no. 128207, doi: 10.1016/j.neucom.2024.128207.

[78] M. A. Islam, M. Ismail, O. Boyaci, R. Atat, and S. Shannigrahi, "Graph neural network based prediction of data traffic in cyber-physical smart power grids," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids (SmartGridComm)*, Oct. 2022, pp. 322–327, doi: 10.1109/SmartGridComm52983.2022.9960963.

[79] S. R. Fahim, R. Atat, C. Kececi, A. Takiddin, M. Ismail, K. R. Davis, and E. Serpedin, "Graph autoencoder-based power attacks detection for resilient electrified transportation systems," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 4, pp. 9539–9553, Dec. 2024, doi: 10.1109/TTE.2024.3355094.

[80] C. He, M. Liu, S. M. Hsiang, and N. Pierce, "Synthesizing ontology and graph neural network to unveil the implicit rules for us bridge preservation decisions," *J. Manage. Eng.*, vol. 40, no. 3, 2024, Art. no. 04024007, doi: 10.1061/JMENEA.MEENG-5803.

[81] F. Liu and I. L. Al-Qadi, "Graph neural network-based 3D asphalt concrete pavement response modelling," *Int. J. Pavement Eng.*, vol. 26, no. 1, Dec. 2025, Art. no. 2464201, doi: 10.1080/10298436.2025.2464201.

[82] L. Yu, B. Du, X. Hu, L. Sun, L. Han, and W. Lv, "Deep spatio-temporal graph convolutional network for traffic accident prediction," *Neurocomputing*, vol. 423, pp. 135–147, Jan. 2021, doi: 10.1016/j.neucom.2020.09.043.

[83] Y. Wu, H.-N. Dai, and H. Tang, "Graph neural networks for anomaly detection in industrial Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9214–9231, Jun. 2022, doi: 10.1109/JIOT.2021.3094295.

[84] Y. Wang, X. Lin, J. Wu, A. K. Bashir, W. Yang, J. Li, and M. Imran, "Contrastive GNN-based traffic anomaly analysis against imbalanced dataset in IoT-based ITS," in *Proc. IEEE Global Commun. Conf.*, Dec. 2022, pp. 3557–3562, doi: 10.1109/GLOBECOM48099.2022.10001621.

[85] W. Liang, Y. Li, K. Xie, D. Zhang, K.-C. Li, A. Souri, and K. Li, "Spatial–temporal aware inductive graph neural network for C-ITS data recovery," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8431–8442, Aug. 2023, doi: 10.1109/TITS.2022.3156266.

[86] X. Huang, S. Hu, W. Wang, I. Kaparias, S. Zhong, X. Na, M. G. Bell, and D.-H. Lee, "Identifying critical links in urban transportation networks based on spatio-temporal dependency learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5583–5597, Jun. 2024, doi: 10.1109/TITS.2023.3339507.

[87] S. Yang, B. Yang, Z. Kang, and L. Deng, "IHG-MA: Inductive heterogeneous graph multi-agent reinforcement learning for multi-intersection traffic signal control," *Neural Netw.*, vol. 139, pp. 265–277, Jul. 2021, doi: 10.1016/j.neunet.2021.03.015.

[88] J. A. Guzmán, G. Pizarro, and F. Núñez, "A reinforcement learning-based distributed control scheme for cooperative intersection traffic control," *IEEE Access*, vol. 11, pp. 57037–57045, 2023, doi: 10.1109/ACCESS.2023.3283218.

[89] T. Wang, Z. Zhu, J. Zhang, J. Tian, and W. Zhang, "A large-scale traffic signal control algorithm based on multi-layer graph deep reinforcement learning," *Transp. Res. C, Emerg. Technol.*, vol. 162, May 2024, Art. no. 104582, doi: 10.1016/j.trc.2024.104582.

[90] S. Yang, Y. Zhang, X. Lu, W. Guo, and H. Miao, "Multi-agent deep reinforcement learning based decision support model for resilient community post-hazard recovery," *Rel. Eng. Syst. Saf.*, vol. 242, Feb. 2024, Art. no. 109754, doi: 10.1016/j.ress.2023.109754.

[91] E. S. da Silva, H. Pedrini, and A. L. D. Santos, "Applying graph neural networks to support decision making on collective intelligent transportation systems," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 4, pp. 4085–4096, Dec. 2023, doi: 10.1109/TNSM.2023.3257993.

[92] A. Zanfei, A. Menapace, B. M. Brentan, M. Righetti, and M. Herrera, "Novel approach for burst detection in water distribution systems based on graph neural networks," *Sustain. Cities Soc.*, vol. 86, Nov. 2022, Art. no. 104090, doi: 10.1016/j.scs.2022.104090.

[93] G. Ö. Garöarsson, F. Boem, and L. Toni, "Graph-based learning for leak detection and localisation in water distribution Networks*," *IFAC-PapersOnLine*, vol. 55, no. 6, pp. 661–666, 2022, doi: 10.1016/j.ifacol.2022.07.203.

[94] W. Wu, X. Pan, Y. Kang, Y. Xu, and L. Han, "To feel the spatial: Graph neural network-based method for leakage risk assessment in water distribution networks," *Water*, vol. 16, no. 14, p. 2017, Jul. 2024, doi: 10.3390/w16142017.

[95] B. Kerimov, R. Taormina, and F. Tscheikner-Gratl, "Towards transferable metamodels for water distribution systems with edge-based graph neural networks," *Water Res.*, vol. 261, Sep. 2024, Art. no. 121933, doi: 10.1016/j.watres.2024.121933.

[96] I. Ashraf, J. Strotherm, L. Hermes, and B. Hammer, "Physics-informed graph neural networks for water distribution systems," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 21905–21913, doi: 10.1609/aaai.v38i20.30192.

[97] L. Xing and L. Sela, "Graph neural networks for state estimation in water distribution systems: Application of supervised and semisupervised learning," *J. Water Resour. Planning Manage.*, vol. 148, no. 5, May 2022, Art. no. 04022018, doi: 10.1061/(asce)wr.1943-5452.0001550.

[98] I. Ashraf, L. Hermes, A. Artelt, and B. Hammer, "Spatial graph convolution neural networks for water distribution systems," in *Proc. Int. Symp. Intell. Data Anal.*, 2023, pp. 29–41, doi: 10.1007/978-3-031-30047-9_3.

[99] H. Truong, A. Tello, A. Lazovik, and V. Degeler, "Graph neural networks for pressure estimation in water distribution systems," *Water Resour. Res.*, vol. 60, no. 7, Jul. 2024, Art. no. e2023WR03674, doi: 10.1029/2023wr036741.

[100] Z. Li, H. Liu, C. Zhang, and G. Fu, "Gated graph neural networks for identifying contamination sources in water distribution systems," *J. Environ. Manage.*, vol. 351, Feb. 2024, Art. no. 119806, doi: 10.1016/j.jenvman.2023.119806.

[101] Z. Li, H. Liu, C. Zhang, and G. Fu, "Real-time water quality prediction in water distribution networks using graph neural networks with sparse monitoring data," *Water Res.*, vol. 250, Feb. 2024, Art. no. 121018, doi: 10.1016/j.watres.2023.121018.

[102] Z. Zhang, W. Tian, C. Lu, Z. Liao, and Z. Yuan, "Graph neural network-based surrogate modelling for real-time hydraulic prediction of urban drainage networks," *Water Res.*, vol. 263, Oct. 2024, Art. no. 122142, doi: 10.1016/j.watres.2024.122142.

[103] G. Moraitis and C. Makropoulos, "A hybrid dynamic graph neural network framework for real-time anomaly detection," *J. Hydroinformatics*, vol. 26, no. 12, pp. 3172–3191, Dec. 2024, doi: 10.2166/hydro.2024.164.

[104] Y. Xu and Z. He, "Pipe failure prediction in the water distribution system using a deep graph convolutional network and temporal failure series," *ACS EST Eng.*, vol. 4, no. 9, pp. 2252–2262, Sep. 2024, doi: 10.1021/acsestengg.4c00234.

[105] T. Mu, C. Zhang, M. Huang, B. Ning, and J. Wang, "Partitioning leakage detection in water distribution systems: A specialized deep learning framework enhanced by spatial–temporal graph convolutional networks," *ACS EST Water*, vol. 4, no. 8, pp. 3453–3463, Aug. 2024, doi: 10.1021/acsestwater.4c00285.

[106] M. Yan and Z. Wang, "Water quality prediction method based on reinforcement learning graph neural network," *IEEE Access*, vol. 12, pp. 184421–184430, 2024, doi: 10.1109/ACCESS.2024.3509744.

[107] S. Peng, J. Cheng, X. Wu, X. Fang, and Q. Wu, "Pressure sensor placement in water supply network based on graph neural network clustering method," *Water*, vol. 14, no. 2, p. 150, Jan. 2022, doi: 10.3390/w14020150.

[108] A. Sirothia and D. S. Narasimhan, "Sensor placement in water distribution networks using graph neural networks," *IFAC-PapersOnLine*, vol. 57, pp. 268–273, 2024, doi: 10.1016/j.ifacol.2024.05.046.

[109] X. Zhou, X. Guan, D. Sun, H. Jiang, J. Peng, Y. Jin, and Y. Zhang, "Transient stability assessment based on gated graph neural network with imbalanced data in Internet of Energy," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9320–9331, Jun. 2022, doi: 10.1109/JIOT.2021.3127895.

[110] Y. Zhang, P. M Karve, and S. Mahadevan, "Graph neural networks for power grid operational risk assessment," 2023, *arXiv:2311.03661*.

[111] Y. Zhu, Y. Zhou, W. Wei, and N. Wang, "Cascading failure analysis based on a physics-informed graph neural network," *IEEE Trans. Power Syst.*, vol. 38, no. 4, pp. 3632–3641, Mar. 2022, doi: 10.1109/TPWRS.2022.3205043.

[112] S. Chadaga, X. Wu, and E. Modiano, "Power failure cascade prediction using graph neural networks," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids (SmartGridComm)*, Oct. 2023, pp. 1–7, doi: 10.1109/smartgridcomm57358.2023.10333943.

[113] P. Gautam, A. Sreejith, and B. Natarajan, "A transductive graph neural network learning for grid resilience analysis," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids (SmartGridComm)*, Oct. 2023, pp. 1–6, doi: 10.1109/smartgridcomm57358.2023.10333912.

[114] D. M. Reddy, D. Dwivedi, P. K. Yemula, and M. Pal, "Data-driven approach to form energy-resilient microgrids with identification of vulnerable nodes in active electrical distribution network," *Int. J. Data Sci. Analytics*, vol. 20, no. 2, pp. 1–12, Aug. 2025, doi: 10.1007/s41060-023-00430-8.

[115] D. Owerko, F. Gama, and A. Ribeiro, "Predicting power outages using graph neural networks," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 743–747, doi: 10.1109/GLOBALSIP.2018.8646486.

[116] Y. Chen, R. A. Jacob, Y. R. Gel, J. Zhang, and H. V. Poor, "Learning power grid outages with higher-order topological neural networks," *IEEE Trans. Power Syst.*, vol. 39, no. 1, pp. 720–732, Jan. 2024, doi: 10.1109/TPWRS.2023.3266956.

[117] A. Varbella, B. Gjorgiev, and G. Sansavini, "Geometric deep learning for online prediction of cascading failures in power grids," *Rel. Eng. Syst. Saf.*, vol. 237, Sep. 2023, Art. no. 109341, doi: 10.1016/j.ress.2023.109341.

[118] O. Boyaci, A. Umunnakwe, A. Sahu, M. R. Narimani, M. Ismail, K. R. Davis, and E. Serpedin, "Graph neural networks based detection of stealth false data injection attacks in smart grids," *IEEE Syst. J.*, vol. 16, no. 2, pp. 2946–2957, Jun. 2022, doi: 10.1109/JSYST.2021.3109082.

[119] A. Presekal, A. Ştefanov, V. S. Rajkumar, and P. Palensky, "Attack graph model for cyber-physical power systems using hybrid deep learning," *IEEE Trans. Smart Grid*, vol. 14, no. 5, pp. 4007–4020, Sep. 2023, doi: 10.1109/TSG.2023.3237011.

[120] S. Peng, M. Liu, L. Chai, and R. Deng, "DST-GNN: A dynamic spatiotemporal graph neural network for cyberattack detection in grid-tied photovoltaic systems," *IEEE Trans. Smart Grid*, vol. 16, no. 1, pp. 330–343, Jan. 2025, doi: 10.1109/TSG.2024.3445113.

[121] S. H. Haghshenas, M. A. Hasnat, and M. Naeini, "A temporal graph neural network for cyber attack detection and localization in smart grids," in *Proc. IEEE Power Energy Soc. Innov. Smart Grid Technol. Conf. (ISGT)*, Jan. 2023, pp. 1–5, doi: 10.1109/ISGT51731.2023.10066446.

[122] S. P. Nandanoori, S. Guan, S. Kundu, S. Pal, K. Agarwal, Y. Wu, and S. Choudhury, "Graph neural network and Koopman models for learning networked dynamics: A comparative study on power grid transients prediction," *IEEE Access*, vol. 10, pp. 32337–32349, 2022, doi: 10.1109/ACCESS.2022.3160710.

[123] Q. Yang and A. Sadeghi, "Graph convolutional networks for power system state estimation," in *Proc. IEEE Smartgridcom Conf.*, 2020.

[124] O. Kundacina, M. Cosovic, D. Miskovic, and D. Vukobratovic, "Distributed nonlinear state estimation in electric power systems using graph neural networks," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids (SmartGridComm)*, Oct. 2022, pp. 8–13, doi: 10.1109/SmartGridComm52983.2022.9960967.

[125] R. Madbhavi, B. Natarajan, and B. Srinivasan, "Graph neural network-based distribution system state estimators," *IEEE Trans. Ind. Informat.*, vol. 19, no. 12, pp. 11630–11639, Dec. 2023, doi: 10.1109/TII.2023.3248082.

[126] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, and T. Ngo, "State estimation for power distribution system using graph neural networks," in *Proc. IEEE Electric Ship Technol. Symp. (ESTS)*, Aug. 2023, pp. 441–446, doi: 10.1109/ests56571.2023.10220523.

[127] J. Hu, Q. Wang, Y. Ye, and Y. Tang, "High-resolution real-time power systems state estimation: A combined physics-embedded and data-driven perspective," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1532–1544, Mar. 2025, doi: 10.1109/TPWRS.2024.3447783.

[128] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Appl. Energy*, vol. 358, Mar. 2024, Art. no. 122602, doi: 10.1016/j.apenergy.2023.122602.

[129] B. Habib, E. Isufi, W. van Breda, A. Jongepier, and J. L. Cremer, "Deep statistical solver for distribution system state estimation," *IEEE Trans. Power Syst.*, vol. 39, no. 2, pp. 4039–4050, Mar. 2024, doi: 10.1109/TPWRS.2023.3290358.

[130] G. Lu and S. Bu, "Advanced probabilistic transient stability assessment for operational planning: A physics-informed graphical learning approach," *IEEE Trans. Power Syst.*, vol. 40, no. 1, pp. 740–752, Jan. 2025, doi: 10.1109/TPWRS.2024.3406674.

[131] Q. Chen, S. Bu, H. Wang, and C. Lei, "Real-time multi-stability risk assessment and visualization of power systems: A graph neural network-based method," *IEEE Trans. Power Syst.*, vol. 40, no. 4, pp. 2955–2968, Jul. 2025, doi: 10.1109/TPWRS.2024.3524406.

[132] K. Rusek, P. Boryło, P. Jaglarz, F. Geyer, A. Cabellos, and P. Chołda, "RiskNet: Neural risk assessment in networks of unreliable resources," *J. Netw. Syst. Manage.*, vol. 31, no. 3, p. 64, Jul. 2023, doi: 10.1007/s10922-023-09755-y.

[133] A. Ahmed, S. Basumallik, A. Gholami, S. K. Sadanandan, M. H. N. Namaki, A. K. Srivastava, and Y. Wu, "Spatio-temporal deep graph network for event detection, localization, and classification in cyber-physical electric distribution system," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 2397–2407, Feb. 2024, doi: 10.1109/TII.2023.3290942.

[134] A. Yaniv, P. Kumar, and Y. Beck, "Towards adoption of GNNs for power flow applications in distribution systems," *Electric Power Syst. Res.*, vol. 216, Mar. 2023, Art. no. 109005, doi: 10.1016/j.epsr.2022.109005.

[135] Y. Pei, J. Yang, J. Wang, P. Xu, T. Zhou, and F. Wu, "An emergency control strategy for undervoltage load shedding of power system: A graph deep reinforcement learning method," *IET Gener., Transmiss. Distribution*, vol. 17, no. 9, pp. 2130–2141, May 2023, doi: 10.1049/gtd2.12795.

[136] Y.-Y. Fu and H.-D. Chiang, "Toward optimal multi-period network reconfiguration for increasing the hosting capacity of distribution networks," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2017, pp. 1–5, doi: 10.1109/PESGM.2017.8274614.

[137] B. Chen, C. Chen, J. Wang, and K. L. Butler-Purry, "Sequential service restoration for unbalanced distribution systems and microgrids," *IEEE Trans. Power Syst.*, vol. 33, no. 2, pp. 1507–1520, Mar. 2018, doi: 10.1109/TPWRS.2017.2720122.

[138] T. Zhao and J. Wang, "Learning sequential distribution system restoration via graph-reinforcement learning," *IEEE Trans. Power Syst.*, vol. 37, no. 2, pp. 1601–1611, Mar. 2022, doi: 10.1109/TPWRS.2021.3102870.
[139] B. Zhang, D. Cao, W. Hu, A. M. Y. M. Ghias, and Z. Chen, "Physics-informed multi-agent deep reinforcement learning enabled distributed voltage control for active distribution network using PV inverters," *Int. J. Electr. Power Energy Syst.*, vol. 155, Jan. 2024, Art. no. 109641, doi: 10.1016/j.ijepes.2023.109641.
[140] X. Wang and N. Lu, "Alleviating imbalanced problems of reinforcement learning when applying in real-time power network dispatching and control," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124730, doi: 10.1016/j.eswa.2024.124730.
[141] Q. Xing, Z. Chen, T. Zhang, X. Li, and K. Sun, "Real-time optimal scheduling for active distribution networks: A graph reinforcement learning method," *Int. J. Electr. Power Energy Syst.*, vol. 145, Feb. 2023, Art. no. 108637, doi: 10.1016/j.ijepes.2022.108637.
[142] J. B. Hansen, S. N. Anfinsen, and F. M. Bianchi, "Power flow balancing with decentralized graph neural networks," *IEEE Trans. Power Syst.*, vol. 38, no. 3, pp. 2423–2433, May 2023, doi: 10.1109/TPWRS.2022.3195301.
[143] S. S. Bhavanasi, L. Pappone, and F. Esposito, "Dealing with changes: Resilient routing via graph neural networks and multi-agent deep reinforcement learning," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 3, pp. 2283–2294, Sep. 2023, doi: 10.1109/TNSM.2023.3287936.
[144] A. Swaminathan, M. Chaba, D. K. Sharma, and U. Ghosh, "GraphNET: Graph neural networks for routing optimization in software defined networks," *Comput. Commun.*, vol. 178, pp. 169–182, Oct. 2021, doi: 10.1016/j.comcom.2021.07.025.
[145] T. Zhu, X. Chen, L. Chen, W. Wang, and G. Wei, "GCLR: GNN-based cross layer optimization for multipath TCP by routing," *IEEE Access*, vol. 8, pp. 17060–17070, 2020, doi: 10.1109/ACCESS.2020.2966045.
[146] J. Guo and C. Yang, "Learning power control for cellular systems with heterogeneous graph neural network," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2021, pp. 1–6, doi: 10.1109/WCNC49053.2021.9417260.
[147] K. Hou, Q. Xu, X. Zhang, Y. Huang, and L. Yang, "User association and power allocation based on unsupervised graph model in ultra-dense network," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2021, pp. 1–6, doi: 10.1109/WCNC49053.2021.9417279.
[148] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 101–115, Jan. 2021, doi: 10.1109/JSAC.2020.3036965.
[149] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "A graph neural network approach for scalable wireless power control," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2019, pp. 1–6, doi: 10.1109/GCWkshps45667.2019.9024538.
[150] N. Naderializadeh, M. Eisen, and A. Ribeiro, "Wireless power control via counterfactual optimization of graph neural networks," in *Proc. IEEE 21st Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, May 2020, pp. 1–5, doi: 10.1109/SPAWC48557.2020.9154336.
[151] M. Lee, G. Yu, and G. Y. Li, "Graph embedding-based wireless link scheduling with few training samples," *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, pp. 2282–2294, Apr. 2021, doi: 10.1109/TWC.2020.3040983.
[152] F. Geyer, "DeepComNet: Performance evaluation of network topologies using graph-based deep learning," *Perform. Eval.*, vol. 130, pp. 1–16, Apr. 2019, doi: 10.1016/j.peva.2018.12.003.
[153] J. Mao, L. Cao, C. Gao, H. Wang, H. Fan, D. Jin, and Y. Li, "Detecting vulnerable nodes in urban infrastructure interdependent network," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2023, pp. 4617–4627, doi: 10.1145/3580305.3599804.
[154] J. Wu and P. Wang, "Generative design for resilience of interdependent network systems," *J. Mech. Design*, vol. 145, no. 3, Mar. 2023, Art. no. 031705, doi: 10.1115/1.4056078.
[155] L. Rodriguez-Garcia, M. M. Hosseini, T. M. Mosier, and M. Parvania, "Resilience analytics for interdependent power and water distribution systems," *IEEE Trans. Power Syst.*, vol. 37, no. 6, pp. 4244–4257, Nov. 2022, doi: 10.1109/TPWRS.2022.3149463.
[156] X. Ding, H. Wang, X. Zhang, C. Ma, and H.-F. Zhang, "Dual nature of cyber–physical power systems and the mitigation strategies," *Rel. Eng. Syst. Saf.*, vol. 244, Apr. 2024, Art. no. 109958, doi: 10.1016/j.ress.2024.109958.
[157] F. Mogyorósi, P. Revisnyei, and A. Pašić, "Modeling and upgrade of disaster-resilient interdependent networks using machine learning," *Opt. Switching Netw.*, vol. 55, Jan. 2025, Art. no. 100791, doi: 10.1016/j.osn.2024.100791.
[158] A. Takiddin, R. Atat, M. Ismail, O. Boyaci, K. R. Davis, and E. Serpedin, "Generalized graph neural network-based detection of false data injection attacks in smart grids," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 3, pp. 618–630, Jun. 2023, doi: 10.1109/TETCI.2022.3232821.

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